

## JESS Thermodynamic Database v8.9

## Chemical species in reactions with Holmium

24-Jan-23

| Charge                                                                                                                                                                             | CAS        | Count | Molecular formula  | Mol. mass |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|--------------------|-----------|
| <b>[12]N4:Acet*4-4</b>                                                                                                                                                             |            |       |                    |           |
| -4                                                                                                                                                                                 | ---        | 116   | C(16)H(24)N(4)O(8) | 400.389   |
| <b>[14]N2O3:MeCOO*2-2</b>                                                                                                                                                          |            |       |                    |           |
| -2                                                                                                                                                                                 | ---        | 62    | C(14)H(24)N(2)O(7) | 332.354   |
| <b>[15]N2O3</b><br>1,4,10-Trioxa-7,13-diazacyclopentadecane; Kryptofix 21; Cryptand 2,1; 4,7,13-Trioxa-1,10-diazacyclopentadecane; Trioxa(2,1) cryptand                            |            |       |                    |           |
| 0                                                                                                                                                                                  | 31249-95-3 | 31    | C(10)H(22)N(2)O(3) | 218.296   |
| <b>[18]N2O4:EtCOO*2-2</b><br>1,10-Diaza-4,7,13,16-tetraoxacyclooctadecane-N,N'-di-b-propionate; K22DP; oddp                                                                        |            |       |                    |           |
| -2                                                                                                                                                                                 | ---        | 23    | C(18)H(32)N(2)O(8) | 404.461   |
| <b>[18]N2O4:MeCOO-1</b><br>1,10-Diaza-4,7,13,16-tetraoxacyclooctadecane-N-acetate ion; K22MA                                                                                       |            |       |                    |           |
| -1                                                                                                                                                                                 | ---        | 21    | C(14)H(27)N(2)O(6) | 319.378   |
| <b>[18]N2O4:MeCOO*2-2</b>                                                                                                                                                          |            |       |                    |           |
| -2                                                                                                                                                                                 | ---        | 43    | C(16)H(28)N(2)O(8) | 376.407   |
| <b>[18]N6</b><br>1,4,7,10,13,16-Hexaazacyclooctadecane; [18]aneN6; 18-Azacrown-6; Hexacyclen; A6-18-Crown-6; A6-18C6; [18]azacoronared-6; 18-membered macrocyclic hexamine         |            |       |                    |           |
| 0                                                                                                                                                                                  | 296-35-5   | 66    | C(12)H(30)N(6)     | 258.410   |
| <b>[2.1.1]crypt</b><br>[2.1.1]crypt; 4,7,13,18-Tetraoxa-1,10-diazabicyclo[8.5.5]eicosane; Kryptofix 211; C211; 1,10-Diaza-4,7,13,18-tetraoxabicyclo[8.5.5]eicosane; Cryptand 2,1,1 |            |       |                    |           |
| 0                                                                                                                                                                                  | 31250-06-3 | 37    | C(14)H(28)N(2)O(4) | 288.387   |
| <b>[2.2.1]crypt</b>                                                                                                                                                                |            |       |                    |           |

|                                                                                                                                                                                               |            |    |                    |         |
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| [2.2.1]cryptand; 4,7,13,16,21-Pentaoxa-1,10-diazabicyclo[8.8.5]tricosane; 1,10-Diaza-4,7,13,16,21-pentaoxabicyclo[8.8.5]tricosane; Cryptand 2,2,1                                             |            |    |                    |         |
| 0                                                                                                                                                                                             | 31364-42-8 | 39 | C(16)H(32)N(2)O(5) | 332.441 |
| <b>[2.2.2]crypt</b><br>[2.2.2]cryptand; 4,7,13,16,21,24-Hexaoxa-1,10-diazabicyclo[8.8.8]hexacosane; Kryptofix 222; Cryptand 2,2,2; 1,10-Diaza-4,7,13,16,21,24-hexaoxabicyclo[8.8.8]hexacosane |            |    |                    |         |
| 0                                                                                                                                                                                             | 23978-09-8 | 48 | C(18)H(36)N(2)O(6) | 376.494 |
| <b>[9]N3:Acet*3-3</b>                                                                                                                                                                         |            |    |                    |         |
| -3                                                                                                                                                                                            | ---        | 45 | C(12)H(18)N(3)O(6) | 300.291 |
| <b>[Hex]14DiCOO-2</b><br>Cyclohexane-trans-1,4-dicarboxylate ion                                                                                                                              |            |    |                    |         |
| -2                                                                                                                                                                                            | ---        | 16 | C(8)H(10)O(4)      | 170.165 |
| <b>[Pent]11DiCOO-2</b>                                                                                                                                                                        |            |    |                    |         |
| -2                                                                                                                                                                                            | ---        | 27 | C(7)H(8)O(4)       | 156.138 |
| <b>[Pent]11OHCOO-1</b>                                                                                                                                                                        |            |    |                    |         |
| -1                                                                                                                                                                                            | ---        | 69 | C(6)H(9)O(3)       | 129.136 |
| <b>11DiMeEDTA-4</b>                                                                                                                                                                           |            |    |                    |         |
| -4                                                                                                                                                                                            | ---        | 29 | C(12)H(16)N(2)O(8) | 316.268 |
| <b>12BisCOOMeAmEthan-2</b>                                                                                                                                                                    |            |    |                    |         |
| -2                                                                                                                                                                                            | ---        | 35 | C(6)H(10)N(2)O(4)  | 174.156 |
| <b>12BisCOOMeOxEthan-2</b>                                                                                                                                                                    |            |    |                    |         |
| -2                                                                                                                                                                                            | ---        | 63 | C(6)H(8)O(6)       | 176.126 |
| <b>12PrDiol</b><br>1,2-Propanediol; Propylene Glycol; Propane-1,2-diol                                                                                                                        |            |    |                    |         |
| 0                                                                                                                                                                                             | 57-55-6    | 26 | C(3)H(8)O(2)       | 76.0953 |
| <b>1BuEDTA-4</b><br>dl-(1-Butylethylene)dinitrilotetraacetate                                                                                                                                 |            |    |                    |         |
| -4                                                                                                                                                                                            | ---        | 28 | C(14)H(20)N(2)O(8) | 344.321 |

|                                                                          |     |     |                           |         |
|--------------------------------------------------------------------------|-----|-----|---------------------------|---------|
| <b>1COOPrAmMalon-3</b>                                                   |     |     |                           |         |
| -3                                                                       | --- | 27  | C (7) H (8) N (1) O (6)   | 202.144 |
| <b>1NO2Naphthol-1</b>                                                    |     |     |                           |         |
| -1                                                                       | --- | 15  | C (10) H (6) N (1) O (2)  | 172.163 |
| <b>1OH47DiSulf2Naphthoic-4</b><br>1-Hydroxy-4,7-disulfo-2-naphthoate ion |     |     |                           |         |
| -4                                                                       | --- | 74  | C (11) H (4) O (9) S (2)  | 344.267 |
| <b>1OH4Sulf2Naphthoic-3</b>                                              |     |     |                           |         |
| -3                                                                       | --- | 111 | C (11) H (5) O (6) S (1)  | 265.217 |
| <b>1PrEDTA-4</b><br>DL- (1-Propylethylene) dinitrilotetraacetate         |     |     |                           |         |
| -4                                                                       | --- | 31  | C (13) H (18) N (2) O (6) | 298.296 |
| <b>22PrNTA-3</b><br>IPNTA ion                                            |     |     |                           |         |
| -3                                                                       | --- | 37  | C (9) H (12) N (1) O (6)  | 230.197 |
| <b>234TriOHPentDioic-2</b>                                               |     |     |                           |         |
| -2                                                                       | --- | 39  | C (5) H (6) O (7)         | 178.098 |
| <b>23DiOH2MeButan-1</b><br>2,3-Dihydroxy-2-methylbutanate ion            |     |     |                           |         |
| -1                                                                       | --- | 66  | C (5) H (9) O (4)         | 133.124 |
| <b>23DiOH2MePropan-1</b><br>2,3-Dihydroxy-2-methylpropanoic acid         |     |     |                           |         |
| -1                                                                       | --- | 55  | C (4) H (7) O (4)         | 119.097 |
| <b>23DiOH2OHMePropan-1</b>                                               |     |     |                           |         |
| -1                                                                       | --- | 46  | C (4) H (7) O (5)         | 135.097 |
| <b>23DiOH6SulfNaphthoic-3</b>                                            |     |     |                           |         |
| -3                                                                       | --- | 84  | C (10) H (5) O (5) S (1)  | 237.207 |

|                                                                             |           |     |                    |         |
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| <b>26DiCOOPiperNAcet-3</b>                                                  |           |     |                    |         |
| -3                                                                          | ---       | 45  | C(9)H(10)N(1)O(6)  | 228.182 |
| <b>26PyridineDiCOO-2</b><br>Dipicolinate ion; 2,6-Pyridinedicarboxylate ion |           |     |                    |         |
| -2                                                                          | ---       | 143 | C(7)H(3)N(1)O(4)   | 165.105 |
| <b>2AmBenz-1</b>                                                            |           |     |                    |         |
| -1                                                                          | 1462-61-9 | 107 | C(7)H(6)N(1)O(2)   | 136.130 |
| <b>2Amphenol-1</b><br>o-Aminophenolate ion                                  |           |     |                    |         |
| -1                                                                          | ---       | 39  | C(6)H(6)N(1)O(1)   | 108.120 |
| <b>2BenzNTA-3</b>                                                           |           |     |                    |         |
| -3                                                                          | ---       | 29  | C(13)H(12)N(1)O(6) | 278.241 |
| <b>2EtNTA-3</b><br>PNTA ion                                                 |           |     |                    |         |
| -3                                                                          | ---       | 39  | C(8)H(10)N(1)O(6)  | 216.171 |
| <b>2HexNTA-3</b>                                                            |           |     |                    |         |
| -3                                                                          | ---       | 25  | C(12)H(18)N(1)O(6) | 272.278 |
| <b>2MeBuDioic-2</b><br>Methylsuccinate (2-Methylbutanedioate)               |           |     |                    |         |
| -2                                                                          | ---       | 30  | C(5)H(6)O(4)       | 130.100 |
| <b>2MeNTA-3</b><br>MNTA anion; MGDA anion                                   |           |     |                    |         |
| -3                                                                          | ---       | 48  | C(7)H(8)N(1)O(6)   | 202.144 |
| <b>2MePrEDTA-4</b>                                                          |           |     |                    |         |
| -4                                                                          | ---       | 30  | C(14)H(20)N(2)O(8) | 344.321 |
| <b>2NO1Naphthol-1</b>                                                       |           |     |                    |         |
| -1                                                                          | ---       | 42  | C(10)H(6)N(1)O(2)  | 172.163 |

|                                                   |     |     |                               |         |
|---------------------------------------------------|-----|-----|-------------------------------|---------|
| <b>2OH2Et3MeButan-1</b>                           |     |     |                               |         |
| -1                                                | --- | 31  | C (7) H (13) O (3)            | 145.178 |
| <b>2OH2EtButan-1</b>                              |     |     |                               |         |
| -1                                                | --- | 39  | C (6) H (11) O (3)            | 131.152 |
| <b>2OH2MeButan-1</b>                              |     |     |                               |         |
| -1                                                | --- | 56  | C (5) H (9) O (3)             | 117.125 |
| <b>2OH2MePropan-1</b>                             |     |     |                               |         |
| -1                                                | --- | 197 | C (4) H (7) O (3)             | 103.098 |
| <b>2OHPentan-1</b>                                |     |     |                               |         |
| -1                                                | --- | 15  | C (5) H (9) O (3)             | 117.125 |
| <b>2OHTriMDTA-4</b>                               |     |     |                               |         |
| -4                                                | --- | 57  | C (11) H (14) N (2) O (9)     | 318.240 |
| <b>2PrEDTA-4</b>                                  |     |     |                               |         |
| -4                                                | --- | 30  | C (13) H (18) N (2) O (8)     | 330.295 |
| <b>2PrNTA-3</b><br>BNTA ion                       |     |     |                               |         |
| -3                                                | --- | 37  | C (9) H (12) N (1) O (6)      | 230.197 |
| <b>2Thiobarb-2</b><br>Thiobarbiturate ion         |     |     |                               |         |
| -2                                                | --- | 38  | C (4) H (2) N (2) O (2) S (1) | 142.132 |
| <b>35DiNitrSal-2</b><br>3,5-Dinitrosalicylate ion |     |     |                               |         |
| -2                                                | --- | 65  | C (7) H (2) N (2) O (7)       | 226.102 |
| <b>35DiSulfSal-4</b>                              |     |     |                               |         |
| -4                                                | --- | 56  | C (7) H (2) O (9) S (2)       | 294.208 |
| <b>3Br5SulfSal-3</b>                              |     |     |                               |         |

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|-------------------------------------------------------------------------------------------------------|------------|-----|-------------------------|---------|
| -3                                                                                                    | ---        | 89  | C(7)H(2)Br(1)O(6)S(1)   | 294.053 |
| <b>3FBenz-1</b>                                                                                       |            |     |                         |         |
| -1                                                                                                    | ---        | 14  | C(7)H(4)F(1)O(2)        | 139.106 |
| <b>3Furoic-1</b>                                                                                      |            |     |                         |         |
| -1                                                                                                    | ---        | 14  | C(5)H(3)O(3)            | 111.077 |
| <b>3OH7Sulf2Naphthoic-3</b>                                                                           |            |     |                         |         |
| -3                                                                                                    | ---        | 32  | C(11)H(5)O(6)S(1)       | 265.217 |
| <b>3OHBenz-2</b><br>3-Hydroxybenzoate anion                                                           |            |     |                         |         |
| -2                                                                                                    | ---        | 17  | C(7)H(4)O(3)            | 136.107 |
| <b>3OHPropan-1</b><br>3-Hydroxypropanoate ion                                                         |            |     |                         |         |
| -1                                                                                                    | ---        | 42  | C(3)H(5)O(3)            | 89.0709 |
| <b>4OHBenz-2</b><br>4-Hydroxybenzoate anion                                                           |            |     |                         |         |
| -2                                                                                                    | ---        | 20  | C(7)H(4)O(3)            | 136.107 |
| <b>5ATP-4</b><br>Adenosine-5'-(tetrahydrogentriphosphate) ion; ATP ion; Adenosine-5'-triphosphate ion |            |     |                         |         |
| -4                                                                                                    | 13265-06-0 | 481 | C(10)H(12)N(5)O(13)P(3) | 503.153 |
| <b>6BrKoj-1</b><br>6-Bromokojate ion                                                                  |            |     |                         |         |
| -1                                                                                                    | ---        | 19  | C(6)H(4)Br(1)O(4)       | 219.999 |
| <b>6IKoj-1</b><br>6-Iodokojate ion                                                                    |            |     |                         |         |
| -1                                                                                                    | ---        | 18  | C(6)H(4)I(1)O(4)        | 266.995 |
| <b>Acac-1</b><br>Acetylacetonate ion                                                                  |            |     |                         |         |
| -1                                                                                                    | 17272-66-1 | 267 | C(5)H(7)O(2)            | 99.1094 |

|                                                                                                                       |            |      |                    |         |
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| <b>Acetic-1</b><br>Acetate ion                                                                                        |            |      |                    |         |
| -1                                                                                                                    | 71-50-1    | 558  | C(2)H(3)O(2)       | 59.0446 |
| <b>Ala-1</b><br>Alanine ion; L-Alanine ion; L-2-Aminopropanoate ion                                                   |            |      |                    |         |
| -1                                                                                                                    | ---        | 802  | C(3)H(6)N(1)O(2)   | 88.0861 |
| <b>AlaGly-1</b>                                                                                                       |            |      |                    |         |
| -1                                                                                                                    | ---        | 44   | C(5)H(9)N(2)O(3)   | 145.138 |
| <b>AlizarinRedS-3</b>                                                                                                 |            |      |                    |         |
| -3                                                                                                                    | ---        | 52   | C(14)H(5)O(7)S(1)  | 317.250 |
| <b>Asp-2</b><br>Aspartate ion; Aminosuccinate ion; Asparagete ion; 2-Aminobutanedioate ion; L-2-Aminobutanedioate ion |            |      |                    |         |
| -2                                                                                                                    | 63-71-8    | 776  | C(4)H(5)N(1)O(4)   | 131.088 |
| <b>Atrolactic-1</b>                                                                                                   |            |      |                    |         |
| -1                                                                                                                    | ---        | 66   | C(9)H(9)O(3)       | 165.169 |
| <b>BEATA-4</b><br>BEATA anion                                                                                         |            |      |                    |         |
| -4                                                                                                                    | ---        | 18   | C(18)H(21)N(3)O(8) | 407.380 |
| <b>Benzoic-1</b><br>Benzoate ion; Benzenecarboxylate ion                                                              |            |      |                    |         |
| -1                                                                                                                    | 766-76-7   | 208  | C(7)H(5)O(2)       | 121.116 |
| <b>Benzoylacetone-1</b>                                                                                               |            |      |                    |         |
| -1                                                                                                                    | ---        | 122  | C(10)H(9)O(2)      | 161.180 |
| <b>Br-1</b><br>Bromide ion                                                                                            |            |      |                    |         |
| -1                                                                                                                    | 24959-67-9 | 1213 | Br(1)              | 79.9040 |
| <b>Br2(1)</b><br>Bromine, liquid                                                                                      |            |      |                    |         |

|                                                                                                                           |            |      |                            |         |
|---------------------------------------------------------------------------------------------------------------------------|------------|------|----------------------------|---------|
| 0                                                                                                                         | ---        | 285  | Br (2)                     | 159.808 |
| <b>CarbMeAsp-3</b>                                                                                                        |            |      |                            |         |
| -3                                                                                                                        | ---        | 37   | C (6) H (6) N (1) O (6)    | 188.117 |
| <b>Cat-2</b><br>Catecholate ion; 1,2-Dihydroxybenzoate ion; 1,2-Benzenedioate ion; Pyrocatecholate ion; Oxyphenoxide      |            |      |                            |         |
| -2                                                                                                                        | 19021-48-8 | 333  | C (6) H (4) O (2)          | 108.097 |
| <b>CDTA-4</b><br>trans-1,2-cyclohexylenedinitrilotetraacetate ion; trans-1,2-diaminocyclohexanetetraacetate ion; CDTA ion |            |      |                            |         |
| -4                                                                                                                        | 22005-54-5 | 301  | C (14) H (18) N (2) O (8)  | 342.306 |
| <b>CEDTA-5</b>                                                                                                            |            |      |                            |         |
| -5                                                                                                                        | ---        | 22   | C (15) H (20) N (3) O (10) | 402.338 |
| <b>Citric-3</b><br>Citrate ion                                                                                            |            |      |                            |         |
| -3                                                                                                                        | 126-44-3   | 1347 | C (6) H (5) O (7)          | 189.102 |
| <b>Cl-1</b><br>Chloride ion                                                                                               |            |      |                            |         |
| -1                                                                                                                        | 16887-00-6 | 2242 | Cl (1)                     | 35.4530 |
| <b>Cl2 (g)</b><br>Chlorine gas                                                                                            |            |      |                            |         |
| 0                                                                                                                         | 7782-50-5  | 589  | Cl (2)                     | 70.9060 |
| <b>CO3-2</b><br>Carbonate ion                                                                                             |            |      |                            |         |
| -2                                                                                                                        | 3812-32-6  | 1435 | C (1) O (3)                | 60.0092 |
| <b>CPDTA-4</b>                                                                                                            |            |      |                            |         |
| -4                                                                                                                        | ---        | 37   | C (13) H (16) N (2) O (8)  | 328.279 |
| <b>CPTA-4</b><br>Cyclopentane-cis-1,2,3,4-tetracarboxylate anion; Cis-1,2,3,4-cyclopentanetetetracarboxylate anion        |            |      |                            |         |
| -4                                                                                                                        | ---        | 21   | C (9) H (6) O (8)          | 242.142 |



|                                                                                                                                                                                      |             |     |                         |         |
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| <b>CrO4-2</b><br>Chromate ion                                                                                                                                                        |             |     |                         |         |
| -2                                                                                                                                                                                   | 11104-59-9  | 127 | Cr(1)O(4)               | 115.994 |
| <b>Croconic-2</b><br>Croconate ion                                                                                                                                                   |             |     |                         |         |
| -2                                                                                                                                                                                   | ---         | 37  | C(2)O(5)                | 104.019 |
| <b>Cu+2_HOEDTA-3</b>                                                                                                                                                                 |             |     |                         |         |
| -1                                                                                                                                                                                   | ---         | 22  | C(10)H(15)Cu(1)N(2)O(7) | 338.784 |
| <b>Cu+2_Tren</b>                                                                                                                                                                     |             |     |                         |         |
| 2                                                                                                                                                                                    | ---         | 20  | C(6)H(18)Cu(1)N(4)      | 209.782 |
| <b>Cys-2</b><br>Cysteinate ion; L-Cysteinate ion; beta-mercaptoalanate ion; 2-amino-3-mercaptopropanoate ion; 2-amino-3-mercaptopropionate ion; alpha-amino-beta-thiolpropionate ion |             |     |                         |         |
| -2                                                                                                                                                                                   | 104170-20-9 | 673 | C(3)H(5)N(1)O(2)S(1)    | 119.138 |
| <b>DEATA-4</b><br>DEATA anion                                                                                                                                                        |             |     |                         |         |
| -4                                                                                                                                                                                   | ---         | 19  | C(14)H(21)N(3)O(8)      | 359.336 |
| <b>DETAP-4</b>                                                                                                                                                                       |             |     |                         |         |
| -4                                                                                                                                                                                   | ---         | 36  | C(13)H(18)N(2)O(9)      | 346.294 |
| <b>DHEGly-1</b>                                                                                                                                                                      |             |     |                         |         |
| -1                                                                                                                                                                                   | ---         | 121 | C(6)H(12)N(1)O(4)       | 162.166 |
| <b>Di(2MeAc)EDDA-4</b><br>Ethylenedinitrilo-N,N'-di(2-methylacetic)-N,N'-diacetate                                                                                                   |             |     |                         |         |
| -4                                                                                                                                                                                   | ---         | 34  | C(12)H(16)N(2)O(8)      | 316.268 |
| <b>DiBuMalon-2</b>                                                                                                                                                                   |             |     |                         |         |
| -2                                                                                                                                                                                   | ---         | 33  | C(11)H(18)O(4)          | 214.262 |
| <b>Dien</b><br>Diethylenetriamine; Iminobis(2-ethylamine); Dien; 1,4,7-Triazaheptane; 3-Azapentane-                                                                                  |             |     |                         |         |

|                                                                                                                                                                       |          |      |                     |          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------|---------------------|----------|
| 1,5-diamine; N1-(2-Aminoethyl)-1,2-ethanediamine; 3NH-pd; 2,2-tri; 2,2'-Diaminodiethylamine; 2,2'-Iminobis(ethylamine)                                                |          |      |                     |          |
| 0                                                                                                                                                                     | 111-40-0 | 254  | C(4)H(13)N(3)       | 103.167  |
| <b>DiEtMalon-2</b><br>Diethylmalonate ion                                                                                                                             |          |      |                     |          |
| -2                                                                                                                                                                    | ---      | 57   | C(7)H(10)O(4)       | 158.154  |
| <b>Diglycol-2</b><br>Diglycolate ion                                                                                                                                  |          |      |                     |          |
| -2                                                                                                                                                                    | ---      | 130  | C(4)H(4)O(5)        | 132.073  |
| <b>DiPrMalon-2</b>                                                                                                                                                    |          |      |                     |          |
| -2                                                                                                                                                                    | ---      | 41   | C(9)H(14)O(4)       | 186.208  |
| <b>Dithizone-1</b>                                                                                                                                                    |          |      |                     |          |
| -1                                                                                                                                                                    | ---      | 74   | C(13)H(11)N(4)S(1)  | 255.317  |
| <b>dlBDTA-4</b><br>DL-2,3-Butylenedinitrilotetraacetate ion; DL-1,2-Dimethylethylenedinitrilotetraacetate ion; dlBDTA anion                                           |          |      |                     |          |
| -4                                                                                                                                                                    | ---      | 48   | C(12)H(16)N(2)O(8)  | 316.268  |
| <b>DMPA-1</b><br>DMPA ion                                                                                                                                             |          |      |                     |          |
| -1                                                                                                                                                                    | ---      | 36   | C(5)H(9)O(4)        | 133.124  |
| <b>DTPA-5</b><br>Diethylenetrinitrilotetraacetate ion; 1,4,7-triazaheptane-1,1,7,7-pentaacetate ion; N,N-bis[2-[bis(carboxymethyl)amino]ethyl]glycinate ion; DTPA ion |          |      |                     |          |
| -5                                                                                                                                                                    | ---      | 343  | C(14)H(18)N(3)O(10) | 388.311  |
| <b>DTTAP-5</b>                                                                                                                                                        |          |      |                     |          |
| -5                                                                                                                                                                    | ---      | 33   | C(15)H(20)N(3)O(10) | 402.338  |
| <b>e-1</b><br>Electron                                                                                                                                                |          |      |                     |          |
| -1                                                                                                                                                                    | ---      | 3399 | E(1)                | 0.000000 |
| <b>EDD3Me2BuDA-4</b>                                                                                                                                                  |          |      |                     |          |
| -4                                                                                                                                                                    | ---      | 28   | C(16)H(24)N(2)O(8)  | 372.375  |

|                                                                                                                                                                                                      |          |      |                        |         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------|------------------------|---------|
| <b>EDDA-2</b><br>Ethylenediiminodiacetate ion; Ethylenediamine-N,N'-diacetate ion; N,N'-ethylene-diglycinate ion; EDDA ion; N,N'-1,2-ethanediylbisglycinate ion; N,N'-Ethylenediaminediacetate anion |          |      |                        |         |
| -2                                                                                                                                                                                                   | ---      | 154  | C(6)H(10)N(2)O(4)      | 174.156 |
| <b>EDDG-4</b><br>EDDG; Ethylenediiminodi-2-pentanedioate ion                                                                                                                                         |          |      |                        |         |
| -4                                                                                                                                                                                                   | ---      | 70   | C(12)H(16)N(2)O(8)     | 316.268 |
| <b>EDDM-4</b><br>Ethylenediamine-N,N'-dimalonate ion                                                                                                                                                 |          |      |                        |         |
| -4                                                                                                                                                                                                   | ---      | 64   | C(8)H(8)N(2)O(8)       | 260.160 |
| <b>EDDPentDA-4</b>                                                                                                                                                                                   |          |      |                        |         |
| -4                                                                                                                                                                                                   | ---      | 29   | C(16)H(24)N(2)O(8)     | 372.375 |
| <b>EDDPrDA-4</b><br>Ethylenediamine-N,N'-diacetate-N,N'-dipropionate; Ethylenedinitrilo-N,N'-di(3-propanoate)-N,N'-diacetate                                                                         |          |      |                        |         |
| -4                                                                                                                                                                                                   | ---      | 53   | C(12)H(16)N(2)O(8)     | 316.268 |
| <b>EDDS-4</b><br>Ethylenediamine-N,N'-disuccinate ion                                                                                                                                                |          |      |                        |         |
| -4                                                                                                                                                                                                   | ---      | 90   | C(10)H(12)N(2)O(8)     | 288.214 |
| <b>EDTA-4</b><br>Ethylenediaminetetraacetate anion                                                                                                                                                   |          |      |                        |         |
| -4                                                                                                                                                                                                   | ---      | 1457 | C(10)H(12)N(2)O(8)     | 288.214 |
| <b>EDTMP-8</b><br>Ethylenediaminetetramethylphosphonate ion                                                                                                                                          |          |      |                        |         |
| -8                                                                                                                                                                                                   | ---      | 134  | C(6)H(12)N(2)O(12)P(4) | 428.064 |
| <b>EEDTA-4</b>                                                                                                                                                                                       |          |      |                        |         |
| -4                                                                                                                                                                                                   | ---      | 77   | C(12)H(16)N(2)O(9)     | 332.267 |
| <b>EGTA-4</b><br>Ethylenebis(oxyethylenenitrilo)tetraacetate ion; EGTA ion                                                                                                                           |          |      |                        |         |
| -4                                                                                                                                                                                                   | 370-65-0 | 132  | C(14)H(20)N(2)O(10)    | 376.320 |

|                                                                                                                                          |            |      |                    |         |
|------------------------------------------------------------------------------------------------------------------------------------------|------------|------|--------------------|---------|
| <b>EtDiAm</b><br>Ethylenediamine; 1,2-Ethanediamine; 1,2-Diaminoethane; en !                                                             |            |      |                    |         |
| 0                                                                                                                                        | 107-15-3   | 879  | C(2)H(8)N(2)       | 60.0989 |
| <b>EtEDTA-4</b>                                                                                                                          |            |      |                    |         |
| -4                                                                                                                                       | ---        | 34   | C(12)H(16)N(2)O(8) | 316.268 |
| <b>EtGlycol</b><br>Ethylene glycol; Ethanediol; Ethane-1,2-diol                                                                          |            |      |                    |         |
| 0                                                                                                                                        | 107-21-1   | 26   | C(2)H(6)O(2)       | 62.0684 |
| <b>EtThioAcet-1</b><br>Ethylthioacetate ion                                                                                              |            |      |                    |         |
| -1                                                                                                                                       | ---        | 69   | C(4)H(7)O(2)S(1)   | 119.158 |
| <b>F-1</b><br>Fluoride ion                                                                                                               |            |      |                    |         |
| -1                                                                                                                                       | 16984-48-8 | 1090 | F(1)               | 18.9984 |
| <b>F2 (g)</b><br>Fluorine gas                                                                                                            |            |      |                    |         |
| 0                                                                                                                                        | 7782-41-4  | 259  | F(2)               | 37.9968 |
| <b>Fe+3_CN-1(6)</b><br>Hexacyanoferrate(III) ion; Ferricyanide ion                                                                       |            |      |                    |         |
| -3                                                                                                                                       | ---        | 29   | C(6)Fe(1)N(6)      | 211.951 |
| <b>Fe+3_Ho+3_CN-1(6)</b>                                                                                                                 |            |      |                    |         |
| 0                                                                                                                                        | ---        | 1    | C(6)Fe(1)Ho(1)N(6) | 376.881 |
| <b>Fumaric-2</b><br>Fumarate ion; trans-Butenedioate ion                                                                                 |            |      |                    |         |
| -2                                                                                                                                       | 142-42-7   | 70   | C(4)H(2)O(4)       | 114.058 |
| <b>Glu-2</b><br>Glutamate ion; 2-aminopentanedioate ion; L-Glutamate ion; alpha-aminoglutarate ion; 1-aminopropane-1,3-dicarboxylate ion |            |      |                    |         |
| -2                                                                                                                                       | 138-18-1   | 650  | C(5)H(7)N(1)O(4)   | 145.115 |

|                                                            |            |      |                       |         |
|------------------------------------------------------------|------------|------|-----------------------|---------|
| <b>Gluconic-1</b><br>Gluconate ion                         |            |      |                       |         |
| -1                                                         | ---        | 133  | C(6)H(11)O(7)         | 195.149 |
| <b>Gly-1</b><br>Glycinate ion; Aminoacetate ion            |            |      |                       |         |
| -1                                                         | ---        | 1356 | C(2)H(4)N(1)O(2)      | 74.0593 |
| <b>GlyAla-1</b><br>Glycyl-L-alaninate ion                  |            |      |                       |         |
| -1                                                         | ---        | 101  | C(5)H(9)N(2)O(3)      | 145.138 |
| <b>Glycerol</b><br>Glycerol; Glycerin; Propane-1,2,3-triol |            |      |                       |         |
| 0                                                          | 56-81-5    | 21   | C(3)H(8)O(3)          | 92.0947 |
| <b>Glycolic-1</b><br>Glycolate ion                         |            |      |                       |         |
| -1                                                         | 57122-18-6 | 342  | C(2)H(3)O(3)          | 75.0440 |
| <b>GlyGly-1</b><br>Glycylglycinate ion                     |            |      |                       |         |
| -1                                                         | 23372-53-4 | 314  | C(4)H(7)N(2)O(3)      | 131.111 |
| <b>GlyGlyGly-1</b><br>Glyglycylglycinate ion               |            |      |                       |         |
| -1                                                         | ---        | 91   | C(6)H(10)N(3)O(4)     | 188.163 |
| <b>GlyLeu-1</b><br>Glycylleucinate ion                     |            |      |                       |         |
| -1                                                         | ---        | 92   | C(8)H(15)N(2)O(3)     | 187.219 |
| <b>GlyMet-1</b>                                            |            |      |                       |         |
| -1                                                         | ---        | 40   | C(7)H(13)N(2)O(3)S(1) | 205.252 |
| <b>Glyoxylic-1</b><br>Glyoxylate ion                       |            |      |                       |         |
| -1                                                         | ---        | 73   | C(2)H(1)O(3)          | 73.0281 |

|                                            |            |       |                         |         |
|--------------------------------------------|------------|-------|-------------------------|---------|
| <b>GlyPhe-1</b><br>Glycylphenylalanate ion |            |       |                         |         |
| -1                                         | ---        | 34    | C(11)H(13)N(2)O(3)      | 221.236 |
| <b>GlySer-1</b>                            |            |       |                         |         |
| -1                                         | ---        | 37    | C(5)H(9)N(2)O(4)        | 161.138 |
| <b>H+1</b><br>Hydrogen ion; Proton         |            |       |                         |         |
| 1                                          | 12408-02-5 | 31679 | H(1)                    | 1.00794 |
| <b>H+1_1OH47DiSulf2Naphthoic-4</b>         |            |       |                         |         |
| -3                                         | ---        | 15    | C(11)H(5)O(9)S(2)       | 345.275 |
| <b>H+1_1OH4Sulf2Naphthoic-3</b>            |            |       |                         |         |
| -2                                         | ---        | 16    | C(11)H(6)O(6)S(1)       | 266.225 |
| <b>H+1_2SHAcet-2</b>                       |            |       |                         |         |
| -1                                         | ---        | 42    | C(2)H(3)O(2)S(1)        | 91.1046 |
| <b>H+1_2SHPropan-2</b>                     |            |       |                         |         |
| -1                                         | ---        | 28    | C(3)H(5)O(2)S(1)        | 105.132 |
| <b>H+1_3OH7Sulf2Naphthoic-3</b>            |            |       |                         |         |
| -2                                         | ---        | 9     | C(11)H(6)O(6)S(1)       | 266.225 |
| <b>H+1_3SHPropan-2</b>                     |            |       |                         |         |
| -1                                         | ---        | 26    | C(3)H(5)O(2)S(1)        | 105.132 |
| <b>H+1_5ATP-4</b>                          |            |       |                         |         |
| -3                                         | ---        | 42    | C(10)H(13)N(5)O(13)P(3) | 504.161 |
| <b>H+1_Aluminon-5</b>                      |            |       |                         |         |
| -4                                         | ---        | 20    | C(22)H(10)O(9)          | 418.316 |
| <b>H+1_Ascorbic-2</b>                      |            |       |                         |         |
| -1                                         | ---        | 53    | C(6)H(7)O(6)            | 175.118 |

|                                                                                                                                                                                                            |         |     |                         |         |
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| <b>H+1_Citric-3</b>                                                                                                                                                                                        |         |     |                         |         |
| -2                                                                                                                                                                                                         | ---     | 93  | C(6)H(6)O(7)            | 190.109 |
| <b>H+1_CO3-2</b>                                                                                                                                                                                           |         |     |                         |         |
| -1                                                                                                                                                                                                         | ---     | 253 | C(1)H(1)O(3)            | 61.0171 |
| <b>H+1_DETAP-4</b>                                                                                                                                                                                         |         |     |                         |         |
| -3                                                                                                                                                                                                         | ---     | 18  | C(13)H(19)N(2)O(9)      | 347.302 |
| <b>H+1_DiMeGlyox-2</b>                                                                                                                                                                                     |         |     |                         |         |
| -1                                                                                                                                                                                                         | ---     | 44  | C(4)H(7)N(2)O(2)        | 115.112 |
| <b>H+1_Dithizone-1</b><br>Dithizone; Diphenylthiocarbazone; 1,5-Diphenyl-1,2,4,5-tetraazapent-1-en-3-thione; 1-Phenyl-4-(phenylimino)-3-thio-semicarbazide; 1,5-Diphenyl-3-formazathiol; Dithizon (no 'e') |         |     |                         |         |
| 0                                                                                                                                                                                                          | 60-10-6 | 30  | C(13)H(12)N(4)S(1)      | 256.325 |
| <b>H+1_DTTAP-5</b>                                                                                                                                                                                         |         |     |                         |         |
| -4                                                                                                                                                                                                         | ---     | 16  | C(15)H(21)N(3)O(10)     | 403.346 |
| <b>H+1_EDDPrDA-4</b>                                                                                                                                                                                       |         |     |                         |         |
| -3                                                                                                                                                                                                         | ---     | 16  | C(12)H(17)N(2)O(8)      | 317.276 |
| <b>H+1_HOEDTA-3</b>                                                                                                                                                                                        |         |     |                         |         |
| -2                                                                                                                                                                                                         | ---     | 28  | C(10)H(16)N(2)O(7)      | 276.246 |
| <b>H+1_MeThymolBlue-6</b>                                                                                                                                                                                  |         |     |                         |         |
| -5                                                                                                                                                                                                         | ---     | 20  | C(37)H(39)N(2)O(13)S(1) | 751.782 |
| <b>H+1_N2OHBenzIDA-3</b>                                                                                                                                                                                   |         |     |                         |         |
| -2                                                                                                                                                                                                         | ---     | 34  | C(11)H(11)N(1)O(5)      | 237.212 |
| <b>H+1_NO3-1</b>                                                                                                                                                                                           |         |     |                         |         |
| 0                                                                                                                                                                                                          | ---     | 32  | H(1)N(1)O(3)            | 63.0128 |
| <b>H+1_NTA-3</b>                                                                                                                                                                                           |         |     |                         |         |

|                                                      |         |     |                         |         |
|------------------------------------------------------|---------|-----|-------------------------|---------|
| -2                                                   | ---     | 33  | C(6)H(7)N(1)O(6)        | 189.125 |
| <b>H+1_P3O10-5</b>                                   |         |     |                         |         |
| -4                                                   | ---     | 56  | H(1)O(10)P(3)           | 253.924 |
| <b>H+1_PAN-1</b><br>1-(2-Pyridylazo)-2-naphthol; PAN |         |     |                         |         |
| 0                                                    | 85-85-8 | 9   | C(15)H(11)N(3)O(1)      | 249.272 |
| <b>H+1_PAR-2</b>                                     |         |     |                         |         |
| -1                                                   | ---     | 46  | C(11)H(8)N(3)O(2)       | 214.203 |
| <b>H+1_PO4-3</b><br>Hydrogenphosphate ion            |         |     |                         |         |
| -2                                                   | ---     | 317 | H(1)O(4)P(1)            | 95.9795 |
| <b>H+1_Salicylic-2</b>                               |         |     |                         |         |
| -1                                                   | ---     | 111 | C(7)H(5)O(3)            | 137.115 |
| <b>H+1_SulfSal-3</b>                                 |         |     |                         |         |
| -2                                                   | ---     | 54  | C(7)H(4)O(6)S(1)        | 216.165 |
| <b>H+1_TAR-2</b>                                     |         |     |                         |         |
| -1                                                   | ---     | 23  | C(9)H(6)N(3)O(2)S(1)    | 220.226 |
| <b>H+1_Tiron-4</b>                                   |         |     |                         |         |
| -3                                                   | ---     | 29  | C(6)H(3)O(8)S(2)        | 267.205 |
| <b>H+1(-1)_5ATP-4</b>                                |         |     |                         |         |
| -5                                                   | ---     | 15  | C(10)H(11)N(5)O(13)P(3) | 502.145 |
| <b>H+1(2)_24DHB-3</b>                                |         |     |                         |         |
| -1                                                   | ---     | 27  | C(7)H(5)O(4)            | 153.114 |
| <b>H+1(2)_25DHB-3</b>                                |         |     |                         |         |
| -1                                                   | ---     | 23  | C(7)H(5)O(4)            | 153.114 |
| <b>H+1(2)_5ATP-4</b>                                 |         |     |                         |         |



|                                                                                                                                                                                                                                  |            |      |                                         |         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------|-----------------------------------------|---------|
| -2                                                                                                                                                                                                                               | ---        | 22   | C (10) H (14) N (5) O (13) P (3)        | 505.169 |
| <b>H+1 (2) _Arsenazo1-6</b>                                                                                                                                                                                                      |            |      |                                         |         |
| -4                                                                                                                                                                                                                               | ---        | 19   | C (16) H (9) As (1) N (2) O (11) S (2)  | 544.296 |
| <b>H+1 (2) _HOEDTA-3</b>                                                                                                                                                                                                         |            |      |                                         |         |
| -1                                                                                                                                                                                                                               | ---        | 15   | C (10) H (17) N (2) O (7)               | 277.254 |
| <b>H+1 (2) _MeThymolBlue-6</b>                                                                                                                                                                                                   |            |      |                                         |         |
| -4                                                                                                                                                                                                                               | ---        | 22   | C (37) H (40) N (2) O (13) S (1)        | 752.790 |
| <b>H+1 (2) _P3O10-5 (2)</b>                                                                                                                                                                                                      |            |      |                                         |         |
| -8                                                                                                                                                                                                                               | ---        | 14   | H (2) O (20) P (6)                      | 507.848 |
| <b>H+1 (2) _PO4-3</b><br>Dihydrogenphosphate ion                                                                                                                                                                                 |            |      |                                         |         |
| -1                                                                                                                                                                                                                               | 14066-20-7 | 284  | H (2) O (4) P (1)                       | 96.9875 |
| <b>H+1 (2) _Purpur-3</b>                                                                                                                                                                                                         |            |      |                                         |         |
| -1                                                                                                                                                                                                                               | ---        | 23   | C (8) H (4) N (5) O (6)                 | 266.150 |
| <b>H+1 (3) _HOEDTA-3</b><br>HOEDTA; HEDTA; Versenol; N-(2-Hydroxyethyl)ethylenediaminetriacetic acid; N-Carboxymethyl-N'-(2-hydroxyethyl)-N,N'-ethylenediglycine; HEEDTA; N-(Hydroxyethyl)diaminoethane-N,N',N'-triethanoic acid |            |      |                                         |         |
| 0                                                                                                                                                                                                                                | 150-39-0   | 16   | C (10) H (18) N (2) O (7)               | 278.262 |
| <b>H+1 (3) _MeThymolBlue-6</b>                                                                                                                                                                                                   |            |      |                                         |         |
| -3                                                                                                                                                                                                                               | ---        | 28   | C (37) H (41) N (2) O (13) S (1)        | 753.798 |
| <b>H+1 (3) _Tren</b>                                                                                                                                                                                                             |            |      |                                         |         |
| 3                                                                                                                                                                                                                                | ---        | 18   | C (6) H (21) N (4)                      | 149.260 |
| <b>H+1 (4) _Arsenazo3-8</b>                                                                                                                                                                                                      |            |      |                                         |         |
| -4                                                                                                                                                                                                                               | ---        | 20   | C (22) H (14) As (2) N (4) O (14) S (2) | 772.336 |
| <b>H2 (g)</b><br>Hydrogen gas; Hydrogen                                                                                                                                                                                          |            |      |                                         |         |
| 0                                                                                                                                                                                                                                | 1333-74-0  | 1564 | H (2)                                   | 2.01588 |

|                                 |            |      |                         |         |
|---------------------------------|------------|------|-------------------------|---------|
| <b>H<sub>2</sub>O</b><br>Water  |            |      |                         |         |
| 0                               | 7732-18-5  | 5947 | H(2)O(1)                | 18.0153 |
| <b>HBED-4</b>                   |            |      |                         |         |
| -4                              | ---        | 98   | C(20)H(20)N(2)O(6)      | 384.389 |
| <b>HexEDTA-4</b>                |            |      |                         |         |
| -4                              | ---        | 29   | C(16)H(24)N(2)O(8)      | 372.375 |
| <b>HIMDA-2</b>                  |            |      |                         |         |
| -2                              | ---        | 201  | C(6)H(9)N(1)O(5)        | 175.141 |
| <b>Ho+2</b><br>Holmium(II) ion  |            |      |                         |         |
| 2                               | ---        | 3    | Ho(1)                   | 164.930 |
| <b>Ho+3</b><br>Holmium(III) ion |            |      |                         |         |
| 3                               | 22541-22-6 | 537  | Ho(1)                   | 164.930 |
| <b>Ho+3_[12]N4:Acet*4-4</b>     |            |      |                         |         |
| -1                              | ---        | 1    | C(16)H(24)Ho(1)N(4)O(8) | 565.319 |
| <b>Ho+3_[14]N2O3:MeCOO*2-2</b>  |            |      |                         |         |
| 1                               | ---        | 1    | C(14)H(24)Ho(1)N(2)O(7) | 497.284 |
| <b>Ho+3_[15]N2O3</b>            |            |      |                         |         |
| 3                               | ---        | 1    | C(10)H(22)Ho(1)N(2)O(3) | 383.226 |
| <b>Ho+3_[18]N2O4:EtCOO*2-2</b>  |            |      |                         |         |
| 1                               | ---        | 1    | C(18)H(32)Ho(1)N(2)O(8) | 569.391 |
| <b>Ho+3_[18]N2O4:MeCOO-1</b>    |            |      |                         |         |
| 2                               | ---        | 1    | C(14)H(27)Ho(1)N(2)O(6) | 484.308 |
| <b>Ho+3_[18]N2O4:MeCOO*2-2</b>  |            |      |                         |         |

|                                 |     |   |                                  |         |
|---------------------------------|-----|---|----------------------------------|---------|
| 1                               | --- | 1 | C (16) H (28) Ho (1) N (2) O (8) | 541.337 |
| <b>Ho+3_[18]N6</b>              |     |   |                                  |         |
| 3                               | --- | 1 | C (12) H (30) Ho (1) N (6)       | 423.340 |
| <b>Ho+3_[2.1.1]crypt</b>        |     |   |                                  |         |
| 3                               | --- | 1 | C (14) H (28) Ho (1) N (2) O (4) | 453.317 |
| <b>Ho+3_[2.2.1]crypt</b>        |     |   |                                  |         |
| 3                               | --- | 1 | C (16) H (32) Ho (1) N (2) O (5) | 497.371 |
| <b>Ho+3_[2.2.2]crypt</b>        |     |   |                                  |         |
| 3                               | --- | 1 | C (18) H (36) Ho (1) N (2) O (6) | 541.424 |
| <b>Ho+3_[9]N3:Acet*3-3</b>      |     |   |                                  |         |
| 0                               | --- | 1 | C (12) H (18) Ho (1) N (3) O (6) | 465.221 |
| <b>Ho+3_[Hex]14DiCOO-2</b>      |     |   |                                  |         |
| 1                               | --- | 1 | C (8) H (10) Ho (1) O (4)        | 335.095 |
| <b>Ho+3_[Pent]11DiCOO-2</b>     |     |   |                                  |         |
| 1                               | --- | 1 | C (7) H (8) Ho (1) O (4)         | 321.068 |
| <b>Ho+3_[Pent]11DiCOO-2 (2)</b> |     |   |                                  |         |
| -1                              | --- | 1 | C (14) H (16) Ho (1) O (8)       | 477.206 |
| <b>Ho+3_[Pent]11OHCOO-1</b>     |     |   |                                  |         |
| 2                               | --- | 2 | C (6) H (9) Ho (1) O (3)         | 294.066 |
| <b>Ho+3_[Pent]11OHCOO-1 (2)</b> |     |   |                                  |         |
| 1                               | --- | 2 | C (12) H (18) Ho (1) O (6)       | 423.201 |
| <b>Ho+3_[Pent]11OHCOO-1 (3)</b> |     |   |                                  |         |
| 0                               | --- | 2 | C (18) H (27) Ho (1) O (9)       | 552.337 |
| <b>Ho+3_[Pent]11OHCOO-1 (4)</b> |     |   |                                  |         |
| -1                              | --- | 1 | C (24) H (36) Ho (1) O (12)      | 681.473 |

|                                         |     |   |                                   |         |
|-----------------------------------------|-----|---|-----------------------------------|---------|
| <b>Ho+3_11DiMeEDTA-4</b>                |     |   |                                   |         |
| -1                                      | --- | 1 | C (12) H (16) Ho (1) N (2) O (8)  | 481.198 |
| <b>Ho+3_12BisCOOMeAmEthan-2</b>         |     |   |                                   |         |
| 1                                       | --- | 1 | C (6) H (10) Ho (1) N (2) O (4)   | 339.086 |
| <b>Ho+3_12BisCOOMeAmEthan-2 (2)</b>     |     |   |                                   |         |
| -1                                      | --- | 1 | C (12) H (20) Ho (1) N (4) O (8)  | 513.243 |
| <b>Ho+3_12BisCOOMeOxEthan-2</b>         |     |   |                                   |         |
| 1                                       | --- | 1 | C (6) H (8) Ho (1) O (6)          | 341.056 |
| <b>Ho+3_12BisCOOMeOxEthan-2 (2)</b>     |     |   |                                   |         |
| -1                                      | --- | 1 | C (12) H (16) Ho (1) O (12)       | 517.182 |
| <b>Ho+3_1BuEDTA-4</b>                   |     |   |                                   |         |
| -1                                      | --- | 1 | C (14) H (20) Ho (1) N (2) O (8)  | 509.251 |
| <b>Ho+3_1COOPrAmMalon-3</b>             |     |   |                                   |         |
| 0                                       | --- | 1 | C (7) H (8) Ho (1) N (1) O (6)    | 367.074 |
| <b>Ho+3_1NO2Naphthol-1</b>              |     |   |                                   |         |
| 2                                       | --- | 1 | C (10) H (6) Ho (1) N (1) O (2)   | 337.093 |
| <b>Ho+3_1OH47DiSulf2Naphthoic-4</b>     |     |   |                                   |         |
| -1                                      | --- | 2 | C (11) H (4) Ho (1) O (9) S (2)   | 509.197 |
| <b>Ho+3_1OH47DiSulf2Naphthoic-4 (2)</b> |     |   |                                   |         |
| -5                                      | --- | 2 | C (22) H (8) Ho (1) O (18) S (4)  | 853.465 |
| <b>Ho+3_1OH4Sulf2Naphthoic-3</b>        |     |   |                                   |         |
| 0                                       | --- | 2 | C (11) H (5) Ho (1) O (6) S (1)   | 430.147 |
| <b>Ho+3_1OH4Sulf2Naphthoic-3 (2)</b>    |     |   |                                   |         |
| -3                                      | --- | 2 | C (22) H (10) Ho (1) O (12) S (2) | 695.364 |

|                                     |     |   |                                   |         |
|-------------------------------------|-----|---|-----------------------------------|---------|
| <b>Ho+3_1PrEDTA-4</b>               |     |   |                                   |         |
| -1                                  | --- | 1 | C (13) H (18) Ho (1) N (2) O (6)  | 463.226 |
| <b>Ho+3_22PrNTA-3</b>               |     |   |                                   |         |
| 0                                   | --- | 1 | C (9) H (12) Ho (1) N (1) O (6)   | 395.127 |
| <b>Ho+3_22PrNTA-3 (2)</b>           |     |   |                                   |         |
| -3                                  | --- | 1 | C (18) H (24) Ho (1) N (2) O (12) | 625.325 |
| <b>Ho+3_234TriOHPentDioic-2</b>     |     |   |                                   |         |
| 1                                   | --- | 1 | C (5) H (6) Ho (1) O (7)          | 343.028 |
| <b>Ho+3_23DiOH2MeButan-1</b>        |     |   |                                   |         |
| 2                                   | --- | 2 | C (5) H (9) Ho (1) O (4)          | 298.054 |
| <b>Ho+3_23DiOH2MeButan-1 (2)</b>    |     |   |                                   |         |
| 1                                   | --- | 2 | C (10) H (18) Ho (1) O (8)        | 431.178 |
| <b>Ho+3_23DiOH2MeButan-1 (3)</b>    |     |   |                                   |         |
| 0                                   | --- | 1 | C (15) H (27) Ho (1) O (12)       | 564.302 |
| <b>Ho+3_23DiOH2MePropan-1</b>       |     |   |                                   |         |
| 2                                   | --- | 2 | C (4) H (7) Ho (1) O (4)          | 284.027 |
| <b>Ho+3_23DiOH2MePropan-1 (2)</b>   |     |   |                                   |         |
| 1                                   | --- | 2 | C (8) H (14) Ho (1) O (8)         | 403.124 |
| <b>Ho+3_23DiOH2MePropan-1 (3)</b>   |     |   |                                   |         |
| 0                                   | --- | 1 | C (12) H (21) Ho (1) O (12)       | 522.222 |
| <b>Ho+3_23DiOH2OHMePropan-1</b>     |     |   |                                   |         |
| 2                                   | --- | 1 | C (4) H (7) Ho (1) O (5)          | 300.027 |
| <b>Ho+3_23DiOH2OHMePropan-1 (2)</b> |     |   |                                   |         |
| 1                                   | --- | 1 | C (8) H (14) Ho (1) O (10)        | 435.123 |

|                                        |     |   |                                   |         |
|----------------------------------------|-----|---|-----------------------------------|---------|
| <b>Ho+3_23DiOH2OHMePropan-1 (3)</b>    |     |   |                                   |         |
| 0                                      | --- | 1 | C (12) H (21) Ho (1) O (15)       | 570.220 |
| <b>Ho+3_23DiOH6SulfNaphthoic-3</b>     |     |   |                                   |         |
| 0                                      | --- | 1 | C (10) H (5) Ho (1) O (5) S (1)   | 402.137 |
| <b>Ho+3_23DiOH6SulfNaphthoic-3 (2)</b> |     |   |                                   |         |
| -3                                     | --- | 1 | C (20) H (10) Ho (1) O (10) S (2) | 639.343 |
| <b>Ho+3_26DiCOOPiperNAcet-3</b>        |     |   |                                   |         |
| 0                                      | --- | 2 | C (9) H (10) Ho (1) N (1) O (6)   | 393.112 |
| <b>Ho+3_26DiCOOPiperNAcet-3 (2)</b>    |     |   |                                   |         |
| -3                                     | --- | 1 | C (18) H (20) Ho (1) N (2) O (12) | 621.293 |
| <b>Ho+3_26PyridineDiCOO-2</b>          |     |   |                                   |         |
| 1                                      | --- | 2 | C (7) H (3) Ho (1) N (1) O (4)    | 330.035 |
| <b>Ho+3_26PyridineDiCOO-2 (2)</b>      |     |   |                                   |         |
| -1                                     | --- | 2 | C (14) H (6) Ho (1) N (2) O (8)   | 495.140 |
| <b>Ho+3_26PyridineDiCOO-2 (3)</b>      |     |   |                                   |         |
| -3                                     | --- | 1 | C (21) H (9) Ho (1) N (3) O (12)  | 660.245 |
| <b>Ho+3_2AmBenz-1</b>                  |     |   |                                   |         |
| 2                                      | --- | 2 | C (7) H (6) Ho (1) N (1) O (2)    | 301.060 |
| <b>Ho+3_2AmBenz-1 (2)</b>              |     |   |                                   |         |
| 1                                      | --- | 3 | C (14) H (12) Ho (1) N (2) O (4)  | 437.190 |
| <b>Ho+3_2AmBenz-1 (3)</b>              |     |   |                                   |         |
| 0                                      | --- | 2 | C (21) H (18) Ho (1) N (3) O (6)  | 573.320 |
| <b>Ho+3_2AmBenz-1 (4)</b>              |     |   |                                   |         |
| -1                                     | --- | 1 | C (28) H (24) Ho (1) N (4) O (8)  | 709.451 |

|                             |     |   |                          |         |
|-----------------------------|-----|---|--------------------------|---------|
| <b>Ho+3_2Amphenol-1</b>     |     |   |                          |         |
| 2                           | --- | 2 | C(6)H(6)Ho(1)N(1)O(1)    | 273.050 |
| <b>Ho+3_2Amphenol-1(2)</b>  |     |   |                          |         |
| 1                           | --- | 1 | C(12)H(12)Ho(1)N(2)O(2)  | 381.170 |
| <b>Ho+3_2BenzNTA-3</b>      |     |   |                          |         |
| 0                           | --- | 1 | C(13)H(12)Ho(1)N(1)O(6)  | 443.171 |
| <b>Ho+3_2BenzNTA-3(2)</b>   |     |   |                          |         |
| -3                          | --- | 1 | C(26)H(24)Ho(1)N(2)O(12) | 721.413 |
| <b>Ho+3_2EtNTA-3</b>        |     |   |                          |         |
| 0                           | --- | 1 | C(8)H(10)Ho(1)N(1)O(6)   | 381.101 |
| <b>Ho+3_2EtNTA-3(2)</b>     |     |   |                          |         |
| -3                          | --- | 1 | C(16)H(20)Ho(1)N(2)O(12) | 597.271 |
| <b>Ho+3_2HexNTA-3</b>       |     |   |                          |         |
| 0                           | --- | 1 | C(12)H(18)Ho(1)N(1)O(6)  | 437.208 |
| <b>Ho+3_2MeBuDioic-2</b>    |     |   |                          |         |
| 1                           | --- | 2 | C(5)H(6)Ho(1)O(4)        | 295.030 |
| <b>Ho+3_2MeBuDioic-2(2)</b> |     |   |                          |         |
| -1                          | --- | 1 | C(10)H(12)Ho(1)O(8)      | 425.131 |
| <b>Ho+3_2MeNTA-3</b>        |     |   |                          |         |
| 0                           | --- | 1 | C(7)H(8)Ho(1)N(1)O(6)    | 367.074 |
| <b>Ho+3_2MeNTA-3(2)</b>     |     |   |                          |         |
| -3                          | --- | 1 | C(14)H(16)Ho(1)N(2)O(12) | 569.217 |
| <b>Ho+3_2MePrEDTA-4</b>     |     |   |                          |         |
| -1                          | --- | 1 | C(14)H(20)Ho(1)N(2)O(8)  | 509.251 |

|                                  |     |   |                                  |         |
|----------------------------------|-----|---|----------------------------------|---------|
| <b>Ho+3_2NO1Napthol-1</b>        |     |   |                                  |         |
| 2                                | --- | 1 | C (10) H (6) Ho (1) N (1) O (2)  | 337.093 |
| <b>Ho+3_2NO1Napthol-1 (2)</b>    |     |   |                                  |         |
| 1                                | --- | 1 | C (20) H (12) Ho (1) N (2) O (4) | 509.256 |
| <b>Ho+3_2OH2Et3MeButan-1</b>     |     |   |                                  |         |
| 2                                | --- | 1 | C (7) H (13) Ho (1) O (3)        | 310.108 |
| <b>Ho+3_2OH2Et3MeButan-1 (2)</b> |     |   |                                  |         |
| 1                                | --- | 1 | C (14) H (26) Ho (1) O (6)       | 455.287 |
| <b>Ho+3_2OH2EtButan-1</b>        |     |   |                                  |         |
| 2                                | --- | 1 | C (6) H (11) Ho (1) O (3)        | 296.082 |
| <b>Ho+3_2OH2EtButan-1 (2)</b>    |     |   |                                  |         |
| 1                                | --- | 1 | C (12) H (22) Ho (1) O (6)       | 427.233 |
| <b>Ho+3_2OH2EtButan-1 (3)</b>    |     |   |                                  |         |
| 0                                | --- | 1 | C (18) H (33) Ho (1) O (9)       | 558.385 |
| <b>Ho+3_2OH2EtButan-1 (4)</b>    |     |   |                                  |         |
| -1                               | --- | 1 | C (24) H (44) Ho (1) O (12)      | 689.536 |
| <b>Ho+3_2OH2MeButan-1</b>        |     |   |                                  |         |
| 2                                | --- | 1 | C (5) H (9) Ho (1) O (3)         | 282.055 |
| <b>Ho+3_2OH2MeButan-1 (2)</b>    |     |   |                                  |         |
| 1                                | --- | 1 | C (10) H (18) Ho (1) O (6)       | 399.179 |
| <b>Ho+3_2OH2MeButan-1 (3)</b>    |     |   |                                  |         |
| 0                                | --- | 1 | C (15) H (27) Ho (1) O (9)       | 516.304 |
| <b>Ho+3_2OH2MePropan-1</b>       |     |   |                                  |         |
| 2                                | --- | 2 | C (4) H (7) Ho (1) O (3)         | 268.028 |



|                                |     |   |                                      |         |
|--------------------------------|-----|---|--------------------------------------|---------|
| <b>Ho+3_2OH2MePropan-1 (2)</b> |     |   |                                      |         |
| 1                              | --- | 3 | C (8) H (14) Ho (1) O (6)            | 371.126 |
| <b>Ho+3_2OH2MePropan-1 (3)</b> |     |   |                                      |         |
| 0                              | --- | 3 | C (12) H (21) Ho (1) O (9)           | 474.223 |
| <b>Ho+3_2OH2MePropan-1 (4)</b> |     |   |                                      |         |
| -1                             | --- | 2 | C (16) H (28) Ho (1) O (12)          | 577.321 |
| <b>Ho+3_2OHPentan-1</b>        |     |   |                                      |         |
| 2                              | --- | 1 | C (5) H (9) Ho (1) O (3)             | 282.055 |
| <b>Ho+3_2OHTriMDTA-4</b>       |     |   |                                      |         |
| -1                             | --- | 1 | C (11) H (14) Ho (1) N (2) O (9)     | 483.170 |
| <b>Ho+3_2PrEDTA-4</b>          |     |   |                                      |         |
| -1                             | --- | 1 | C (13) H (18) Ho (1) N (2) O (8)     | 495.225 |
| <b>Ho+3_2PrNTA-3</b>           |     |   |                                      |         |
| 0                              | --- | 1 | C (9) H (12) Ho (1) N (1) O (6)      | 395.127 |
| <b>Ho+3_2PrNTA-3 (2)</b>       |     |   |                                      |         |
| -3                             | --- | 1 | C (18) H (24) Ho (1) N (2) O (12)    | 625.325 |
| <b>Ho+3_2Thiobarb-2</b>        |     |   |                                      |         |
| 1                              | --- | 1 | C (4) H (2) Ho (1) N (2) O (2) S (1) | 307.062 |
| <b>Ho+3_35DiNitrSal-2</b>      |     |   |                                      |         |
| 1                              | --- | 1 | C (7) H (2) Ho (1) N (2) O (7)       | 391.032 |
| <b>Ho+3_35DiSulfSal-4</b>      |     |   |                                      |         |
| -1                             | --- | 2 | C (7) H (2) Ho (1) O (9) S (2)       | 459.138 |
| <b>Ho+3_35DiSulfSal-4 (2)</b>  |     |   |                                      |         |
| -5                             | --- | 2 | C (14) H (4) Ho (1) O (18) S (4)     | 753.345 |

|                                  |     |   |                                          |         |
|----------------------------------|-----|---|------------------------------------------|---------|
| <b>Ho+3_3Br5SulfSal-3</b>        |     |   |                                          |         |
| 0                                | --- | 2 | C (7) H (2) Br (1) Ho (1) O (6) S (1)    | 458.983 |
| <b>Ho+3_3Br5SulfSal-3 (2)</b>    |     |   |                                          |         |
| -3                               | --- | 2 | C (14) H (4) Br (2) Ho (1) O (12) S (2)  | 753.037 |
| <b>Ho+3_3FBenz-1</b>             |     |   |                                          |         |
| 2                                | --- | 1 | C (7) H (4) F (1) Ho (1) O (2)           | 304.036 |
| <b>Ho+3_3Furoic-1</b>            |     |   |                                          |         |
| 2                                | --- | 1 | C (5) H (3) Ho (1) O (3)                 | 276.007 |
| <b>Ho+3_3OH7Sulf2Naphthoic-3</b> |     |   |                                          |         |
| 0                                | --- | 1 | C (11) H (5) Ho (1) O (6) S (1)          | 430.147 |
| <b>Ho+3_3OHBenz-2</b>            |     |   |                                          |         |
| 1                                | --- | 1 | C (7) H (4) Ho (1) O (3)                 | 301.037 |
| <b>Ho+3_3OHPropan-1</b>          |     |   |                                          |         |
| 2                                | --- | 1 | C (3) H (5) Ho (1) O (3)                 | 254.001 |
| <b>Ho+3_4OHBenz-2</b>            |     |   |                                          |         |
| 1                                | --- | 1 | C (7) H (4) Ho (1) O (3)                 | 301.037 |
| <b>Ho+3_5ATP-4</b>               |     |   |                                          |         |
| -1                               | --- | 2 | C (10) H (12) Ho (1) N (5) O (13) P (3)  | 668.083 |
| <b>Ho+3_5ATP-4_EDTA-4</b>        |     |   |                                          |         |
| -5                               | --- | 2 | C (20) H (24) Ho (1) N (7) O (21) P (3)  | 956.297 |
| <b>Ho+3_5ATP-4 (2)</b>           |     |   |                                          |         |
| -5                               | --- | 1 | C (20) H (24) Ho (1) N (10) O (26) P (6) | 1171.24 |
| <b>Ho+3_6BrKoj-1</b>             |     |   |                                          |         |
| 2                                | --- | 1 | C (6) H (4) Br (1) Ho (1) O (4)          | 384.929 |

|                            |     |   |                                   |         |
|----------------------------|-----|---|-----------------------------------|---------|
| <b>Ho+3_6IKoj-1</b>        |     |   |                                   |         |
| 2                          | --- | 1 | C (6) H (4) Ho (1) I (1) O (4)    | 431.925 |
| <b>Ho+3_Acac-1</b>         |     |   |                                   |         |
| 2                          | --- | 2 | C (5) H (7) Ho (1) O (2)          | 264.039 |
| <b>Ho+3_Acac-1_EDTA-4</b>  |     |   |                                   |         |
| -2                         | --- | 2 | C (15) H (19) Ho (1) N (2) O (10) | 552.253 |
| <b>Ho+3_Acac-1 (2)</b>     |     |   |                                   |         |
| 1                          | --- | 2 | C (10) H (14) Ho (1) O (4)        | 363.149 |
| <b>Ho+3_Acac-1 (3)</b>     |     |   |                                   |         |
| 0                          | --- | 1 | C (15) H (21) Ho (1) O (6)        | 462.258 |
| <b>Ho+3_Acetic-1</b>       |     |   |                                   |         |
| 2                          | --- | 2 | C (2) H (3) Ho (1) O (2)          | 223.975 |
| <b>Ho+3_Acetic-1 (2)</b>   |     |   |                                   |         |
| 1                          | --- | 2 | C (4) H (6) Ho (1) O (4)          | 283.019 |
| <b>Ho+3_Acetic-1 (3)</b>   |     |   |                                   |         |
| 0                          | --- | 1 | C (6) H (9) Ho (1) O (6)          | 342.064 |
| <b>Ho+3_Acetic-1 (4)</b>   |     |   |                                   |         |
| -1                         | --- | 1 | C (8) H (12) Ho (1) O (8)         | 401.109 |
| <b>Ho+3_Ala-1</b>          |     |   |                                   |         |
| 2                          | --- | 1 | C (3) H (6) Ho (1) N (1) O (2)    | 253.016 |
| <b>Ho+3_AlaGly-1</b>       |     |   |                                   |         |
| 2                          | --- | 1 | C (5) H (9) Ho (1) N (2) O (3)    | 310.068 |
| <b>Ho+3_AlizarinRedS-3</b> |     |   |                                   |         |
| 0                          | --- | 1 | C (14) H (5) Ho (1) O (7) S (1)   | 482.180 |

|                                 |     |   |                          |         |
|---------------------------------|-----|---|--------------------------|---------|
| <b>Ho+3_AlizarinRedS-3(3)</b>   |     |   |                          |         |
| -6                              | --- | 1 | C(42)H(15)Ho(1)O(21)S(3) | 1116.68 |
| <b>Ho+3_Asp-2</b>               |     |   |                          |         |
| 1                               | --- | 2 | C(4)H(5)Ho(1)N(1)O(4)    | 296.018 |
| <b>Ho+3_Asp-2(2)</b>            |     |   |                          |         |
| -1                              | --- | 2 | C(8)H(10)Ho(1)N(2)O(8)   | 427.106 |
| <b>Ho+3_Atrolactic-1</b>        |     |   |                          |         |
| 2                               | --- | 2 | C(9)H(9)Ho(1)O(3)        | 330.099 |
| <b>Ho+3_Atrolactic-1(2)</b>     |     |   |                          |         |
| 1                               | --- | 3 | C(18)H(18)Ho(1)O(6)      | 495.267 |
| <b>Ho+3_Atrolactic-1(3)</b>     |     |   |                          |         |
| 0                               | --- | 3 | C(27)H(27)Ho(1)O(9)      | 660.436 |
| <b>Ho+3_Atrolactic-1(4)</b>     |     |   |                          |         |
| -1                              | --- | 2 | C(36)H(36)Ho(1)O(12)     | 825.605 |
| <b>Ho+3_BEATA-4</b>             |     |   |                          |         |
| -1                              | --- | 1 | C(18)H(21)Ho(1)N(3)O(8)  | 572.310 |
| <b>Ho+3_Benzoic-1</b>           |     |   |                          |         |
| 2                               | --- | 2 | C(7)H(5)Ho(1)O(2)        | 286.046 |
| <b>Ho+3_Benzoic-1(2)</b>        |     |   |                          |         |
| 1                               | --- | 1 | C(14)H(10)Ho(1)O(4)      | 407.161 |
| <b>Ho+3_Benzoylacetone-1</b>    |     |   |                          |         |
| 2                               | --- | 2 | C(10)H(9)Ho(1)O(2)       | 326.110 |
| <b>Ho+3_Benzoylacetone-1(2)</b> |     |   |                          |         |
| 1                               | --- | 2 | C(20)H(18)Ho(1)O(4)      | 487.291 |

|                                            |     |   |                          |         |
|--------------------------------------------|-----|---|--------------------------|---------|
| <b>Ho+3_Benzoylacetone-1(3)</b>            |     |   |                          |         |
| 0                                          | --- | 1 | C(30)H(27)Ho(1)O(6)      | 648.471 |
| <b>Ho+3_Br-1</b>                           |     |   |                          |         |
| 2                                          | --- | 2 | Br(1)Ho(1)               | 244.834 |
| <b>Ho+3_Br-1(2)</b>                        |     |   |                          |         |
| 1                                          | --- | 2 | Br(2)Ho(1)               | 324.738 |
| <b>Ho+3_Br-1(3)</b>                        |     |   |                          |         |
| 0                                          | --- | 1 | Br(3)Ho(1)               | 404.642 |
| <b>Ho+3_Br-1(3)_(s)</b><br>Holmium bromide |     |   |                          |         |
| 0                                          | --- | 1 | Br(3)Ho(1)               | 404.642 |
| <b>Ho+3_CarbMeAsp-3</b>                    |     |   |                          |         |
| 0                                          | --- | 1 | C(6)H(6)Ho(1)N(1)O(6)    | 353.047 |
| <b>Ho+3_Cat-2</b>                          |     |   |                          |         |
| 1                                          | --- | 1 | C(6)H(4)Ho(1)O(2)        | 273.027 |
| <b>Ho+3_CDTA-4</b>                         |     |   |                          |         |
| -1                                         | --- | 2 | C(14)H(18)Ho(1)N(2)O(8)  | 507.236 |
| <b>Ho+3_CEDTA-5</b>                        |     |   |                          |         |
| -2                                         | --- | 1 | C(15)H(20)Ho(1)N(3)O(10) | 567.268 |
| <b>Ho+3_Citric-3</b>                       |     |   |                          |         |
| 0                                          | --- | 1 | C(6)H(5)Ho(1)O(7)        | 354.032 |
| <b>Ho+3_Citric-3_NTA-3</b>                 |     |   |                          |         |
| -3                                         | --- | 1 | C(12)H(11)Ho(1)N(1)O(13) | 542.148 |
| <b>Ho+3_Cl-1</b>                           |     |   |                          |         |
| 2                                          | --- | 2 | Cl(1)Ho(1)               | 200.383 |

|                                                                     |     |   |                                  |         |
|---------------------------------------------------------------------|-----|---|----------------------------------|---------|
| <b>Ho+3_Cl-1 (2)</b>                                                |     |   |                                  |         |
| 1                                                                   | --- | 3 | Cl (2) Ho (1)                    | 235.836 |
| <b>Ho+3_Cl-1 (3)</b>                                                |     |   |                                  |         |
| 0                                                                   | --- | 2 | Cl (3) Ho (1)                    | 271.289 |
| <b>Ho+3_Cl-1 (3) _ (s)</b><br>Holmium chloride                      |     |   |                                  |         |
| 0                                                                   | --- | 1 | Cl (3) Ho (1)                    | 271.289 |
| <b>Ho+3_Cl-1 (3) _H2O (6) _ (s)</b><br>Holmium chloride hexahydrate |     |   |                                  |         |
| 0                                                                   | --- | 2 | Cl (3) H (12) Ho (1) O (6)       | 379.381 |
| <b>Ho+3_Cl-1 (4)</b>                                                |     |   |                                  |         |
| -1                                                                  | --- | 1 | Cl (4) Ho (1)                    | 306.742 |
| <b>Ho+3_CO3-2</b>                                                   |     |   |                                  |         |
| 1                                                                   | --- | 1 | C (1) Ho (1) O (3)               | 224.939 |
| <b>Ho+3_CO3-2 (2)</b>                                               |     |   |                                  |         |
| -1                                                                  | --- | 1 | C (2) Ho (1) O (6)               | 284.948 |
| <b>Ho+3_CO3-2 (4)</b>                                               |     |   |                                  |         |
| -5                                                                  | --- | 2 | C (4) Ho (1) O (12)              | 404.967 |
| <b>Ho+3_CO3-2 (4) _Oxalic-2</b>                                     |     |   |                                  |         |
| -7                                                                  | --- | 1 | C (6) Ho (1) O (16)              | 492.986 |
| <b>Ho+3_CPDTA-4</b>                                                 |     |   |                                  |         |
| -1                                                                  | --- | 1 | C (13) H (16) Ho (1) N (2) O (8) | 493.209 |
| <b>Ho+3_CPTA-4</b>                                                  |     |   |                                  |         |
| -1                                                                  | --- | 1 | C (9) H (6) Ho (1) O (8)         | 407.072 |
| <b>Ho+3_CrO4-2</b>                                                  |     |   |                                  |         |
| 1                                                                   | --- | 1 | Cr (1) Ho (1) O (4)              | 280.924 |

|                               |     |   |                                       |         |
|-------------------------------|-----|---|---------------------------------------|---------|
| <b>Ho+3_Croconic-2</b>        |     |   |                                       |         |
| 1                             | --- | 1 | C (2) Ho (1) O (5)                    | 268.949 |
| <b>Ho+3_Croconic-2 (2)</b>    |     |   |                                       |         |
| -1                            | --- | 1 | C (4) Ho (1) O (10)                   | 372.968 |
| <b>Ho+3_Cys-2</b>             |     |   |                                       |         |
| 1                             | --- | 1 | C (3) H (5) Ho (1) N (1) O (2) S (1)  | 284.068 |
| <b>Ho+3_Cys-2 (2)</b>         |     |   |                                       |         |
| -1                            | --- | 1 | C (6) H (10) Ho (1) N (2) O (4) S (2) | 403.206 |
| <b>Ho+3_DEATA-4</b>           |     |   |                                       |         |
| -1                            | --- | 1 | C (14) H (21) Ho (1) N (3) O (8)      | 524.266 |
| <b>Ho+3_DETAP-4</b>           |     |   |                                       |         |
| -1                            | --- | 1 | C (13) H (18) Ho (1) N (2) O (9)      | 511.224 |
| <b>Ho+3_DHEGly-1</b>          |     |   |                                       |         |
| 2                             | --- | 2 | C (6) H (12) Ho (1) N (1) O (4)       | 327.096 |
| <b>Ho+3_DHEGly-1 (2)</b>      |     |   |                                       |         |
| 1                             | --- | 2 | C (12) H (24) Ho (1) N (2) O (8)      | 489.261 |
| <b>Ho+3_Di (2MeAc) EDDA-4</b> |     |   |                                       |         |
| -1                            | --- | 1 | C (12) H (16) Ho (1) N (2) O (8)      | 481.198 |
| <b>Ho+3_DiBuMalon-2</b>       |     |   |                                       |         |
| 1                             | --- | 1 | C (11) H (18) Ho (1) O (4)            | 379.192 |
| <b>Ho+3_DiBuMalon-2 (2)</b>   |     |   |                                       |         |
| -1                            | --- | 1 | C (22) H (36) Ho (1) O (8)            | 593.453 |
| <b>Ho+3_Dien</b>              |     |   |                                       |         |
| 3                             | --- | 1 | C (4) H (13) Ho (1) N (3)             | 268.097 |

|                            |     |   |                         |         |
|----------------------------|-----|---|-------------------------|---------|
| <b>Ho+3_Dien(2)</b>        |     |   |                         |         |
| 3                          | --- | 1 | C(8)H(26)Ho(1)N(6)      | 371.265 |
| <b>Ho+3_DiEtMalon-2</b>    |     |   |                         |         |
| 1                          | --- | 1 | C(7)H(10)Ho(1)O(4)      | 323.084 |
| <b>Ho+3_DiEtMalon-2(2)</b> |     |   |                         |         |
| -1                         | --- | 1 | C(14)H(20)Ho(1)O(8)     | 481.238 |
| <b>Ho+3_Diglycol-2</b>     |     |   |                         |         |
| 1                          | --- | 1 | C(4)H(4)Ho(1)O(5)       | 297.003 |
| <b>Ho+3_Diglycol-2(2)</b>  |     |   |                         |         |
| -1                         | --- | 1 | C(8)H(8)Ho(1)O(10)      | 429.076 |
| <b>Ho+3_Diglycol-2(3)</b>  |     |   |                         |         |
| -3                         | --- | 1 | C(12)H(12)Ho(1)O(15)    | 561.148 |
| <b>Ho+3_DiPrMalon-2</b>    |     |   |                         |         |
| 1                          | --- | 1 | C(9)H(14)Ho(1)O(4)      | 351.138 |
| <b>Ho+3_DiPrMalon-2(2)</b> |     |   |                         |         |
| -1                         | --- | 1 | C(18)H(28)Ho(1)O(8)     | 537.346 |
| <b>Ho+3_Dithizone-1</b>    |     |   |                         |         |
| 2                          | --- | 1 | C(13)H(11)Ho(1)N(4)S(1) | 420.247 |
| <b>Ho+3_d1BDTA-4</b>       |     |   |                         |         |
| -1                         | --- | 1 | C(12)H(16)Ho(1)N(2)O(8) | 481.198 |
| <b>Ho+3_DMPA-1</b>         |     |   |                         |         |
| 2                          | --- | 2 | C(5)H(9)Ho(1)O(4)       | 298.054 |
| <b>Ho+3_DMPA-1(2)</b>      |     |   |                         |         |
| 1                          | --- | 1 | C(10)H(18)Ho(1)O(8)     | 431.178 |



|                           |     |   |                                   |         |
|---------------------------|-----|---|-----------------------------------|---------|
| <b>Ho+3_DTPA-5</b>        |     |   |                                   |         |
| -2                        | --- | 2 | C (14) H (18) Ho (1) N (3) O (10) | 553.241 |
| <b>Ho+3_DTTAP-5</b>       |     |   |                                   |         |
| -2                        | --- | 1 | C (15) H (20) Ho (1) N (3) O (10) | 567.268 |
| <b>Ho+3_EDD3Me2BuDA-4</b> |     |   |                                   |         |
| -1                        | --- | 1 | C (16) H (24) Ho (1) N (2) O (8)  | 537.305 |
| <b>Ho+3_EDDA-2</b>        |     |   |                                   |         |
| 1                         | --- | 2 | C (6) H (10) Ho (1) N (2) O (4)   | 339.086 |
| <b>Ho+3_EDDA-2 (2)</b>    |     |   |                                   |         |
| -1                        | --- | 2 | C (12) H (20) Ho (1) N (4) O (8)  | 513.243 |
| <b>Ho+3_EDDG-4</b>        |     |   |                                   |         |
| -1                        | --- | 1 | C (12) H (16) Ho (1) N (2) O (8)  | 481.198 |
| <b>Ho+3_EDDM-4</b>        |     |   |                                   |         |
| -1                        | --- | 1 | C (8) H (8) Ho (1) N (2) O (8)    | 425.090 |
| <b>Ho+3_EDDM-4 (2)</b>    |     |   |                                   |         |
| -5                        | --- | 1 | C (16) H (16) Ho (1) N (4) O (16) | 685.250 |
| <b>Ho+3_EDDPentDA-4</b>   |     |   |                                   |         |
| -1                        | --- | 1 | C (16) H (24) Ho (1) N (2) O (8)  | 537.305 |
| <b>Ho+3_EDDPrDA-4</b>     |     |   |                                   |         |
| -1                        | --- | 1 | C (12) H (16) Ho (1) N (2) O (8)  | 481.198 |
| <b>Ho+3_EDDS-4</b>        |     |   |                                   |         |
| -1                        | --- | 1 | C (10) H (12) Ho (1) N (2) O (8)  | 453.144 |
| <b>Ho+3_EDDS-4 (2)</b>    |     |   |                                   |         |
| -5                        | --- | 1 | C (20) H (24) Ho (1) N (4) O (16) | 741.358 |

|                            |     |    |                                         |         |
|----------------------------|-----|----|-----------------------------------------|---------|
| <b>Ho+3_EDTA-4</b>         |     |    |                                         |         |
| -1                         | --- | 14 | C (10) H (12) Ho (1) N (2) O (8)        | 453.144 |
| <b>Ho+3_EDTA-4_P3O10-5</b> |     |    |                                         |         |
| -6                         | --- | 2  | C (10) H (12) Ho (1) N (2) O (18) P (3) | 706.060 |
| <b>Ho+3_EDTA-4_Tiron-4</b> |     |    |                                         |         |
| -5                         | --- | 1  | C (16) H (14) Ho (1) N (2) O (16) S (2) | 719.341 |
| <b>Ho+3_EDTA-4 (2)</b>     |     |    |                                         |         |
| -5                         | --- | 2  | C (20) H (24) Ho (1) N (4) O (16)       | 741.358 |
| <b>Ho+3_EDTMP-8</b>        |     |    |                                         |         |
| -5                         | --- | 2  | C (6) H (12) Ho (1) N (2) O (12) P (4)  | 592.994 |
| <b>Ho+3_EEDTA-4</b>        |     |    |                                         |         |
| -1                         | --- | 1  | C (12) H (16) Ho (1) N (2) O (9)        | 497.197 |
| <b>Ho+3_EGTA-4</b>         |     |    |                                         |         |
| -1                         | --- | 1  | C (14) H (20) Ho (1) N (2) O (10)       | 541.250 |
| <b>Ho+3_EtDiAm</b>         |     |    |                                         |         |
| 3                          | --- | 2  | C (2) H (8) Ho (1) N (2)                | 225.029 |
| <b>Ho+3_EtDiAm (2)</b>     |     |    |                                         |         |
| 3                          | --- | 2  | C (4) H (16) Ho (1) N (4)               | 285.128 |
| <b>Ho+3_EtDiAm (3)</b>     |     |    |                                         |         |
| 3                          | --- | 2  | C (6) H (24) Ho (1) N (6)               | 345.227 |
| <b>Ho+3_EtDiAm (4)</b>     |     |    |                                         |         |
| 3                          | --- | 1  | C (8) H (32) Ho (1) N (8)               | 405.326 |
| <b>Ho+3_EtEDTA-4</b>       |     |    |                                         |         |
| -1                         | --- | 1  | C (12) H (16) Ho (1) N (2) O (8)        | 481.198 |

|                                    |     |   |                                        |         |
|------------------------------------|-----|---|----------------------------------------|---------|
| <b>Ho+3_EtThioAcet-1</b>           |     |   |                                        |         |
| 2                                  | --- | 2 | C (4) H (7) Ho (1) O (2) S (1)         | 284.088 |
| <b>Ho+3_EtThioAcet-1 (2)</b>       |     |   |                                        |         |
| 1                                  | --- | 1 | C (8) H (14) Ho (1) O (4) S (2)        | 403.247 |
| <b>Ho+3_F-1</b>                    |     |   |                                        |         |
| 2                                  | --- | 1 | F (1) Ho (1)                           | 183.928 |
| <b>Ho+3_F-1_EDTA-4</b>             |     |   |                                        |         |
| -2                                 | --- | 3 | C (10) H (12) F (1) Ho (1) N (2) O (8) | 472.142 |
| <b>Ho+3_F-1 (2)</b>                |     |   |                                        |         |
| 1                                  | --- | 1 | F (2) Ho (1)                           | 202.927 |
| <b>Ho+3_F-1 (2)_EDTA-4</b>         |     |   |                                        |         |
| -3                                 | --- | 2 | C (10) H (12) F (2) Ho (1) N (2) O (8) | 491.141 |
| <b>Ho+3_F-1 (3)_ (s)</b>           |     |   |                                        |         |
| 0                                  | --- | 2 | F (3) Ho (1)                           | 221.925 |
| <b>Ho+3_F-1 (3)_H2O (0.5)_ (s)</b> |     |   |                                        |         |
| 0                                  | --- | 1 | F (3) H (1) Ho (1) O (0.5)             | 230.933 |
| <b>Ho+3_Fumaric-2</b>              |     |   |                                        |         |
| 1                                  | --- | 1 | C (4) H (2) Ho (1) O (4)               | 278.988 |
| <b>Ho+3_Glu-2</b>                  |     |   |                                        |         |
| 1                                  | --- | 1 | C (5) H (7) Ho (1) N (1) O (4)         | 310.045 |
| <b>Ho+3_Gluconic-1</b>             |     |   |                                        |         |
| 2                                  | --- | 1 | C (6) H (11) Ho (1) O (7)              | 360.079 |
| <b>Ho+3_Gluconic-1 (2)</b>         |     |   |                                        |         |
| 1                                  | --- | 3 | C (12) H (22) Ho (1) O (14)            | 555.228 |

|                            |     |   |                                       |         |
|----------------------------|-----|---|---------------------------------------|---------|
| <b>Ho+3_Gly-1</b>          |     |   |                                       |         |
| 2                          | --- | 1 | C (2) H (4) Ho (1) N (1) O (2)        | 238.989 |
| <b>Ho+3_GlyAla-1</b>       |     |   |                                       |         |
| 2                          | --- | 1 | C (5) H (9) Ho (1) N (2) O (3)        | 310.068 |
| <b>Ho+3_Glycolic-1</b>     |     |   |                                       |         |
| 2                          | --- | 2 | C (2) H (3) Ho (1) O (3)              | 239.974 |
| <b>Ho+3_Glycolic-1 (2)</b> |     |   |                                       |         |
| 1                          | --- | 3 | C (4) H (6) Ho (1) O (6)              | 315.018 |
| <b>Ho+3_Glycolic-1 (3)</b> |     |   |                                       |         |
| 0                          | --- | 2 | C (6) H (9) Ho (1) O (9)              | 390.062 |
| <b>Ho+3_Glycolic-1 (4)</b> |     |   |                                       |         |
| -1                         | --- | 1 | C (8) H (12) Ho (1) O (12)            | 465.106 |
| <b>Ho+3_Glycolic-1 (5)</b> |     |   |                                       |         |
| -2                         | --- | 1 | C (10) H (15) Ho (1) O (15)           | 540.150 |
| <b>Ho+3_GlyGly-1</b>       |     |   |                                       |         |
| 2                          | --- | 1 | C (4) H (7) Ho (1) N (2) O (3)        | 296.041 |
| <b>Ho+3_GlyGlyGly-1</b>    |     |   |                                       |         |
| 2                          | --- | 1 | C (6) H (10) Ho (1) N (3) O (4)       | 353.093 |
| <b>Ho+3_GlyLeu-1</b>       |     |   |                                       |         |
| 2                          | --- | 1 | C (8) H (15) Ho (1) N (2) O (3)       | 352.149 |
| <b>Ho+3_GlyMet-1</b>       |     |   |                                       |         |
| 2                          | --- | 1 | C (7) H (13) Ho (1) N (2) O (3) S (1) | 370.182 |
| <b>Ho+3_Glyoxylic-1</b>    |     |   |                                       |         |
| 2                          | --- | 2 | C (2) H (1) Ho (1) O (3)              | 237.958 |

|                                           |     |   |                            |         |
|-------------------------------------------|-----|---|----------------------------|---------|
| <b>Ho+3_Glyoxylic-1(2)</b>                |     |   |                            |         |
| 1                                         | --- | 2 | C(4)H(2)Ho(1)O(6)          | 310.986 |
| <b>Ho+3_Glyoxylic-1(3)</b>                |     |   |                            |         |
| 0                                         | --- | 1 | C(6)H(3)Ho(1)O(9)          | 384.014 |
| <b>Ho+3_GlyPhe-1</b>                      |     |   |                            |         |
| 2                                         | --- | 1 | C(11)H(13)Ho(1)N(2)O(3)    | 386.166 |
| <b>Ho+3_GlySer-1</b>                      |     |   |                            |         |
| 2                                         | --- | 1 | C(5)H(9)Ho(1)N(2)O(4)      | 326.067 |
| <b>Ho+3_H+1_12BisCOOMeOxEthan-2(2)</b>    |     |   |                            |         |
| 0                                         | --- | 1 | C(12)H(17)Ho(1)O(12)       | 518.190 |
| <b>Ho+3_H+1_1OH47DiSulf2Naphthoic-4</b>   |     |   |                            |         |
| 0                                         | --- | 1 | C(11)H(5)Ho(1)O(9)S(2)     | 510.205 |
| <b>Ho+3_H+1_1OH4Sulf2Naphthoic-3</b>      |     |   |                            |         |
| 1                                         | --- | 2 | C(11)H(6)Ho(1)O(6)S(1)     | 431.155 |
| <b>Ho+3_H+1_23DiOH6SulfNaphthoic-3(2)</b> |     |   |                            |         |
| -2                                        | --- | 1 | C(20)H(11)Ho(1)O(10)S(2)   | 640.351 |
| <b>Ho+3_H+1_2SHAcet-2</b>                 |     |   |                            |         |
| 2                                         | --- | 2 | C(2)H(3)Ho(1)O(2)S(1)      | 256.035 |
| <b>Ho+3_H+1_2SHPropan-2</b>               |     |   |                            |         |
| 2                                         | --- | 2 | C(3)H(5)Ho(1)O(2)S(1)      | 270.062 |
| <b>Ho+3_H+1_3Br5SulfSal-3</b>             |     |   |                            |         |
| 1                                         | --- | 1 | C(7)H(3)Br(1)Ho(1)O(6)S(1) | 459.991 |
| <b>Ho+3_H+1_3OH7Sulf2Naphthoic-3</b>      |     |   |                            |         |
| 1                                         | --- | 1 | C(11)H(6)Ho(1)O(6)S(1)     | 431.155 |

|                             |     |   |                              |         |
|-----------------------------|-----|---|------------------------------|---------|
| <b>Ho+3_H+1_3SHPropan-2</b> |     |   |                              |         |
| 2                           | --- | 2 | C(3)H(5)Ho(1)O(2)S(1)        | 270.062 |
| <b>Ho+3_H+1_5ATP-4</b>      |     |   |                              |         |
| 0                           | --- | 1 | C(10)H(13)Ho(1)N(5)O(13)P(3) | 669.091 |
| <b>Ho+3_H+1_Aluminon-5</b>  |     |   |                              |         |
| -1                          | --- | 1 | C(22)H(10)Ho(1)O(9)          | 583.246 |
| <b>Ho+3_H+1_Ascorbic-2</b>  |     |   |                              |         |
| 2                           | --- | 1 | C(6)H(7)Ho(1)O(6)            | 340.048 |
| <b>Ho+3_H+1_CDTA-4</b>      |     |   |                              |         |
| 0                           | --- | 1 | C(14)H(19)Ho(1)N(2)O(8)      | 508.244 |
| <b>Ho+3_H+1_Citric-3</b>    |     |   |                              |         |
| 1                           | --- | 1 | C(6)H(6)Ho(1)O(7)            | 355.039 |
| <b>Ho+3_H+1_Citric-3(2)</b> |     |   |                              |         |
| -2                          | --- | 2 | C(12)H(11)Ho(1)O(14)         | 544.141 |
| <b>Ho+3_H+1_CO3-2</b>       |     |   |                              |         |
| 2                           | --- | 2 | C(1)H(1)Ho(1)O(3)            | 225.947 |
| <b>Ho+3_H+1_DETAP-4</b>     |     |   |                              |         |
| 0                           | --- | 1 | C(13)H(19)Ho(1)N(2)O(9)      | 512.232 |
| <b>Ho+3_H+1_DiMeGlyox-2</b> |     |   |                              |         |
| 2                           | --- | 2 | C(4)H(7)Ho(1)N(2)O(2)        | 280.042 |
| <b>Ho+3_H+1_Dithizone-1</b> |     |   |                              |         |
| 3                           | --- | 1 | C(13)H(12)Ho(1)N(4)S(1)      | 421.255 |
| <b>Ho+3_H+1_DTPA-5</b>      |     |   |                              |         |
| -1                          | --- | 1 | C(14)H(19)Ho(1)N(3)O(10)     | 554.249 |

|                                |     |   |                                         |         |
|--------------------------------|-----|---|-----------------------------------------|---------|
| <b>Ho+3_H+1_DTTAP-5</b>        |     |   |                                         |         |
| -1                             | --- | 1 | C (15) H (21) Ho (1) N (3) O (10)       | 568.276 |
| <b>Ho+3_H+1_EDDPrDA-4</b>      |     |   |                                         |         |
| 0                              | --- | 2 | C (12) H (17) Ho (1) N (2) O (8)        | 482.206 |
| <b>Ho+3_H+1_EDTA-4</b>         |     |   |                                         |         |
| 0                              | --- | 2 | C (10) H (13) Ho (1) N (2) O (8)        | 454.152 |
| <b>Ho+3_H+1_EDTA-4_P3O10-5</b> |     |   |                                         |         |
| -5                             | --- | 2 | C (10) H (13) Ho (1) N (2) O (18) P (3) | 707.068 |
| <b>Ho+3_H+1_EDTMP-8</b>        |     |   |                                         |         |
| -4                             | --- | 2 | C (6) H (13) Ho (1) N (2) O (12) P (4)  | 594.001 |
| <b>Ho+3_H+1_Fumaric-2</b>      |     |   |                                         |         |
| 2                              | --- | 1 | C (4) H (3) Ho (1) O (4)                | 279.995 |
| <b>Ho+3_H+1_HBED-4</b>         |     |   |                                         |         |
| 0                              | --- | 2 | C (20) H (21) Ho (1) N (2) O (6)        | 550.327 |
| <b>Ho+3_H+1_HOEDTA-3 (2)</b>   |     |   |                                         |         |
| -2                             | --- | 1 | C (20) H (31) Ho (1) N (4) O (14)       | 716.415 |
| <b>Ho+3_H+1_IDA-2</b>          |     |   |                                         |         |
| 2                              | --- | 3 | C (4) H (6) Ho (1) N (1) O (4)          | 297.026 |
| <b>Ho+3_H+1_Malonic-2</b>      |     |   |                                         |         |
| 2                              | --- | 1 | C (3) H (3) Ho (1) O (4)                | 267.984 |
| <b>Ho+3_H+1_Malonic-2 (2)</b>  |     |   |                                         |         |
| 0                              | --- | 1 | C (6) H (5) Ho (1) O (8)                | 370.031 |
| <b>Ho+3_H+1_N2OHBenzIDA-3</b>  |     |   |                                         |         |
| 1                              | --- | 1 | C (11) H (11) Ho (1) N (1) O (5)        | 402.142 |

|                              |     |   |                           |         |
|------------------------------|-----|---|---------------------------|---------|
| <b>Ho+3_H+1_NO3-1(4)</b>     |     |   |                           |         |
| 0                            | --- | 2 | H(1)Ho(1)N(4)O(12)        | 413.958 |
| <b>Ho+3_H+1_PAR-2</b>        |     |   |                           |         |
| 2                            | --- | 1 | C(11)H(8)Ho(1)N(3)O(2)    | 379.133 |
| <b>Ho+3_H+1_PO4-3</b>        |     |   |                           |         |
| 1                            | --- | 3 | H(1)Ho(1)O(4)P(1)         | 260.910 |
| <b>Ho+3_H+1_Salicylic-2</b>  |     |   |                           |         |
| 2                            | --- | 2 | C(7)H(5)Ho(1)O(3)         | 302.045 |
| <b>Ho+3_H+1_SulfSal-3</b>    |     |   |                           |         |
| 1                            | --- | 1 | C(7)H(4)Ho(1)O(6)S(1)     | 381.095 |
| <b>Ho+3_H+1_TAR-2</b>        |     |   |                           |         |
| 2                            | --- | 1 | C(9)H(6)Ho(1)N(3)O(2)S(1) | 385.156 |
| <b>Ho+3_H+1_TetMDTA-4</b>    |     |   |                           |         |
| 0                            | --- | 2 | C(12)H(17)Ho(1)N(2)O(8)   | 482.206 |
| <b>Ho+3_H+1_Tiron-4</b>      |     |   |                           |         |
| 0                            | --- | 1 | C(6)H(3)Ho(1)O(8)S(2)     | 432.135 |
| <b>Ho+3_H+1_Tiron-4(2)</b>   |     |   |                           |         |
| -4                           | --- | 1 | C(12)H(5)Ho(1)O(16)S(4)   | 698.332 |
| <b>Ho+3_H+1_TTHA-6</b>       |     |   |                           |         |
| -2                           | --- | 2 | C(18)H(25)Ho(1)N(4)O(12)  | 654.346 |
| <b>Ho+3_H+1(-1)_12PrDiol</b> |     |   |                           |         |
| 2                            | --- | 1 | C(3)H(7)Ho(1)O(2)         | 240.017 |
| <b>Ho+3_H+1(-1)_Citric-3</b> |     |   |                           |         |
| -1                           | --- | 1 | C(6)H(4)Ho(1)O(7)         | 353.024 |



|                                   |     |   |                              |         |
|-----------------------------------|-----|---|------------------------------|---------|
| <b>Ho+3_H+1(-1)_EtGlycol</b>      |     |   |                              |         |
| 2                                 | --- | 1 | C(2)H(5)Ho(1)O(2)            | 225.991 |
| <b>Ho+3_H+1(-1)_Gluconic-1(2)</b> |     |   |                              |         |
| 0                                 | --- | 1 | C(12)H(21)Ho(1)O(14)         | 554.220 |
| <b>Ho+3_H+1(-1)_Glycerol</b>      |     |   |                              |         |
| 2                                 | --- | 1 | C(3)H(7)Ho(1)O(3)            | 256.017 |
| <b>Ho+3_H+1(-1)_OH-1_5ATP-4</b>   |     |   |                              |         |
| -3                                | --- | 1 | C(10)H(12)Ho(1)N(5)O(14)P(3) | 684.082 |
| <b>Ho+3_H+1(-2)_3Br5SulfSal-3</b> |     |   |                              |         |
| -2                                | --- | 1 | C(7)Br(1)Ho(1)O(6)S(1)       | 456.967 |
| <b>Ho+3_H+1(-2)_Gluconic-1(2)</b> |     |   |                              |         |
| -1                                | --- | 1 | C(12)H(20)Ho(1)O(14)         | 553.212 |
| <b>Ho+3_H+1(2)_24DHB-3</b>        |     |   |                              |         |
| 2                                 | --- | 1 | C(7)H(5)Ho(1)O(4)            | 318.044 |
| <b>Ho+3_H+1(2)_25DHB-3</b>        |     |   |                              |         |
| 2                                 | --- | 1 | C(7)H(5)Ho(1)O(4)            | 318.044 |
| <b>Ho+3_H+1(2)_2SHAcet-2(2)</b>   |     |   |                              |         |
| 1                                 | --- | 1 | C(4)H(6)Ho(1)O(4)S(2)        | 347.139 |
| <b>Ho+3_H+1(2)_2SHPropan-2(2)</b> |     |   |                              |         |
| 1                                 | --- | 1 | C(6)H(10)Ho(1)O(4)S(2)       | 375.193 |
| <b>Ho+3_H+1(2)_3SHPropan-2(2)</b> |     |   |                              |         |
| 1                                 | --- | 1 | C(6)H(10)Ho(1)O(4)S(2)       | 375.193 |
| <b>Ho+3_H+1(2)_5ATP-4</b>         |     |   |                              |         |
| 1                                 | --- | 1 | C(10)H(14)Ho(1)N(5)O(13)P(3) | 670.099 |

|                                     |     |   |                                  |         |
|-------------------------------------|-----|---|----------------------------------|---------|
| <b>Ho+3_H+1(2)_Arsenazo1-6</b>      |     |   |                                  |         |
| -1                                  | --- | 1 | C(16)H(9)As(1)Ho(1)N(2)O(11)S(2) | 709.226 |
| <b>Ho+3_H+1(2)_DiMeGlyox-2(2)</b>   |     |   |                                  |         |
| 1                                   | --- | 1 | C(8)H(14)Ho(1)N(4)O(4)           | 395.154 |
| <b>Ho+3_H+1(2)_EDTMP-8</b>          |     |   |                                  |         |
| -3                                  | --- | 2 | C(6)H(14)Ho(1)N(2)O(12)P(4)      | 595.009 |
| <b>Ho+3_H+1(2)_HBED-4</b>           |     |   |                                  |         |
| 1                                   | --- | 1 | C(20)H(22)Ho(1)N(2)O(6)          | 551.335 |
| <b>Ho+3_H+1(2)_HOEDTA-3(2)</b>      |     |   |                                  |         |
| -1                                  | --- | 1 | C(20)H(32)Ho(1)N(4)O(14)         | 717.423 |
| <b>Ho+3_H+1(2)_IDA-2</b>            |     |   |                                  |         |
| 3                                   | --- | 2 | C(4)H(7)Ho(1)N(1)O(4)            | 298.034 |
| <b>Ho+3_H+1(2)_MeThymolBlue-6</b>   |     |   |                                  |         |
| -1                                  | --- | 2 | C(37)H(40)Ho(1)N(2)O(13)S(1)     | 917.720 |
| <b>Ho+3_H+1(2)_N2OHBenzIDA-3(2)</b> |     |   |                                  |         |
| -1                                  | --- | 1 | C(22)H(22)Ho(1)N(2)O(10)         | 639.354 |
| <b>Ho+3_H+1(2)_P3O10-5(2)</b>       |     |   |                                  |         |
| -5                                  | --- | 1 | H(2)Ho(1)O(20)P(6)               | 672.778 |
| <b>Ho+3_H+1(2)_PO4-3</b>            |     |   |                                  |         |
| 2                                   | --- | 2 | H(2)Ho(1)O(4)P(1)                | 261.918 |
| <b>Ho+3_H+1(2)_PO4-3(2)</b>         |     |   |                                  |         |
| -1                                  | --- | 3 | H(2)Ho(1)O(8)P(2)                | 356.889 |
| <b>Ho+3_H+1(2)_Purpur-3</b>         |     |   |                                  |         |
| 2                                   | --- | 1 | C(8)H(4)Ho(1)N(5)O(6)            | 431.080 |

|                                   |     |   |                                   |         |
|-----------------------------------|-----|---|-----------------------------------|---------|
| <b>Ho+3_H+1(2)_Salicylic-2(2)</b> |     |   |                                   |         |
| 1                                 | --- | 2 | C(14)H(10)Ho(1)O(6)               | 439.160 |
| <b>Ho+3_H+1(2)_TetMDTA-4</b>      |     |   |                                   |         |
| 1                                 | --- | 1 | C(12)H(18)Ho(1)N(2)O(8)           | 483.214 |
| <b>Ho+3_H+1(2)_TTHA-6</b>         |     |   |                                   |         |
| -1                                | --- | 1 | C(18)H(26)Ho(1)N(4)O(12)          | 655.354 |
| <b>Ho+3_H+1(3)_EDTMP-8</b>        |     |   |                                   |         |
| -2                                | --- | 1 | C(6)H(15)Ho(1)N(2)O(12)P(4)       | 596.017 |
| <b>Ho+3_H+1(3)_HOEDTA-3(2)</b>    |     |   |                                   |         |
| 0                                 | --- | 1 | C(20)H(33)Ho(1)N(4)O(14)          | 718.430 |
| <b>Ho+3_H+1(3)_MeThymolBlue-6</b> |     |   |                                   |         |
| 0                                 | --- | 2 | C(37)H(41)Ho(1)N(2)O(13)S(1)      | 918.728 |
| <b>Ho+3_H+1(3)_NO3-1(6)</b>       |     |   |                                   |         |
| 0                                 | --- | 1 | H(3)Ho(1)N(6)O(18)                | 539.983 |
| <b>Ho+3_H+1(4)_Arsenazo3-8</b>    |     |   |                                   |         |
| -1                                | --- | 1 | C(22)H(14)As(2)Ho(1)N(4)O(14)S(2) | 937.266 |
| <b>Ho+3_HBED-4</b>                |     |   |                                   |         |
| -1                                | --- | 2 | C(20)H(20)Ho(1)N(2)O(6)           | 549.319 |
| <b>Ho+3_HexEDTA-4</b>             |     |   |                                   |         |
| -1                                | --- | 1 | C(16)H(24)Ho(1)N(2)O(8)           | 537.305 |
| <b>Ho+3_HIMDA-2</b>               |     |   |                                   |         |
| 1                                 | --- | 2 | C(6)H(9)Ho(1)N(1)O(5)             | 340.071 |
| <b>Ho+3_HIMDA-2_HOEDTA-3</b>      |     |   |                                   |         |
| -2                                | --- | 1 | C(16)H(24)Ho(1)N(3)O(12)          | 615.310 |

|                                     |     |   |                                   |         |
|-------------------------------------|-----|---|-----------------------------------|---------|
| <b>Ho+3_HIMDA-2 (2)</b>             |     |   |                                   |         |
| -1                                  | --- | 3 | C (12) H (18) Ho (1) N (2) O (10) | 515.212 |
| <b>Ho+3_HOEDTA-3</b>                |     |   |                                   |         |
| 0                                   | --- | 9 | C (10) H (15) Ho (1) N (2) O (7)  | 440.168 |
| <b>Ho+3_HOEDTA-3_EDTA-4</b>         |     |   |                                   |         |
| -4                                  | --- | 1 | C (20) H (27) Ho (1) N (4) O (15) | 728.382 |
| <b>Ho+3_HOEDTA-3 (2)</b>            |     |   |                                   |         |
| -3                                  | --- | 1 | C (20) H (30) Ho (1) N (4) O (14) | 715.407 |
| <b>Ho+3_HydrazineNNDiAcet-2</b>     |     |   |                                   |         |
| 1                                   | --- | 1 | C (4) H (6) Ho (1) N (2) O (4)    | 311.033 |
| <b>Ho+3_HydrazineNNDiAcet-2 (2)</b> |     |   |                                   |         |
| -1                                  | --- | 1 | C (8) H (12) Ho (1) N (4) O (8)   | 457.135 |
| <b>Ho+3_HydrazineNNDiAcet-2 (3)</b> |     |   |                                   |         |
| -3                                  | --- | 1 | C (12) H (18) Ho (1) N (6) O (12) | 603.238 |
| <b>Ho+3_Hyp-1</b>                   |     |   |                                   |         |
| 2                                   | --- | 1 | C (5) H (8) Ho (1) N (1) O (3)    | 295.053 |
| <b>Ho+3_Hyp-1 (2)</b>               |     |   |                                   |         |
| 1                                   | --- | 1 | C (10) H (16) Ho (1) N (2) O (6)  | 425.177 |
| <b>Ho+3_IButanoic-1</b>             |     |   |                                   |         |
| 2                                   | --- | 2 | C (4) H (7) Ho (1) O (2)          | 252.028 |
| <b>Ho+3_IButanoic-1 (2)</b>         |     |   |                                   |         |
| 1                                   | --- | 2 | C (8) H (14) Ho (1) O (4)         | 339.127 |
| <b>Ho+3_IDA-2</b>                   |     |   |                                   |         |
| 1                                   | --- | 3 | C (4) H (5) Ho (1) N (1) O (4)    | 296.018 |

|                            |     |   |                                   |         |
|----------------------------|-----|---|-----------------------------------|---------|
| <b>Ho+3_IDA-2_EDTA-4</b>   |     |   |                                   |         |
| -3                         | --- | 2 | C (14) H (17) Ho (1) N (3) O (12) | 584.232 |
| <b>Ho+3_IDA-2_HOEDTA-3</b> |     |   |                                   |         |
| -2                         | --- | 1 | C (14) H (20) Ho (1) N (3) O (11) | 571.256 |
| <b>Ho+3_IDA-2 (2)</b>      |     |   |                                   |         |
| -1                         | --- | 2 | C (8) H (10) Ho (1) N (2) O (8)   | 427.106 |
| <b>Ho+3_IDA-2 (3)</b>      |     |   |                                   |         |
| -3                         | --- | 1 | C (12) H (15) Ho (1) N (3) O (12) | 558.194 |
| <b>Ho+3_IDS-4</b>          |     |   |                                   |         |
| -1                         | --- | 1 | C (8) H (7) Ho (1) N (1) O (8)    | 410.076 |
| <b>Ho+3_IO3-1 (3)_(s)</b>  |     |   |                                   |         |
| 0                          | --- | 1 | Ho (1) I (3) O (9)                | 689.625 |
| <b>Ho+3_Itaconic-2</b>     |     |   |                                   |         |
| 1                          | --- | 1 | C (5) H (4) Ho (1) O (4)          | 293.014 |
| <b>Ho+3_Lactic-1</b>       |     |   |                                   |         |
| 2                          | --- | 2 | C (3) H (5) Ho (1) O (3)          | 254.001 |
| <b>Ho+3_Lactic-1 (2)</b>   |     |   |                                   |         |
| 1                          | --- | 3 | C (6) H (10) Ho (1) O (6)         | 343.072 |
| <b>Ho+3_Lactic-1 (3)</b>   |     |   |                                   |         |
| 0                          | --- | 3 | C (9) H (15) Ho (1) O (9)         | 432.143 |
| <b>Ho+3_Lactic-1 (4)</b>   |     |   |                                   |         |
| -1                         | --- | 1 | C (12) H (20) Ho (1) O (12)       | 521.214 |
| <b>Ho+3_LeuGly-1</b>       |     |   |                                   |         |
| 2                          | --- | 1 | C (8) H (15) Ho (1) N (2) O (3)   | 352.149 |

|                            |     |   |                                  |         |
|----------------------------|-----|---|----------------------------------|---------|
| <b>Ho+3_LeuGlyGly-1</b>    |     |   |                                  |         |
| 2                          | --- | 1 | C (10) H (18) Ho (1) N (3) O (4) | 409.201 |
| <b>Ho+3_Maleic-2</b>       |     |   |                                  |         |
| 1                          | --- | 2 | C (4) H (2) Ho (1) O (4)         | 278.988 |
| <b>Ho+3_Maleic-2 (2)</b>   |     |   |                                  |         |
| -1                         | --- | 2 | C (8) H (4) Ho (1) O (8)         | 393.045 |
| <b>Ho+3_Malic-2</b>        |     |   |                                  |         |
| 1                          | --- | 2 | C (4) H (4) Ho (1) O (5)         | 297.003 |
| <b>Ho+3_Malic-2 (2)</b>    |     |   |                                  |         |
| -1                         | --- | 2 | C (8) H (8) Ho (1) O (10)        | 429.076 |
| <b>Ho+3_Malonic-2</b>      |     |   |                                  |         |
| 1                          | --- | 2 | C (3) H (2) Ho (1) O (4)         | 266.977 |
| <b>Ho+3_Malonic-2 (2)</b>  |     |   |                                  |         |
| -1                         | --- | 2 | C (6) H (4) Ho (1) O (8)         | 369.023 |
| <b>Ho+3_Malonic-2 (3)</b>  |     |   |                                  |         |
| -3                         | --- | 1 | C (9) H (6) Ho (1) O (12)        | 471.069 |
| <b>Ho+3_Mandelic-1</b>     |     |   |                                  |         |
| 2                          | --- | 2 | C (8) H (7) Ho (1) O (3)         | 316.072 |
| <b>Ho+3_Mandelic-1 (2)</b> |     |   |                                  |         |
| 1                          | --- | 2 | C (16) H (14) Ho (1) O (6)       | 467.214 |
| <b>Ho+3_mAnisic-1</b>      |     |   |                                  |         |
| 2                          | --- | 1 | C (8) H (7) Ho (1) O (3)         | 316.072 |
| <b>Ho+3_mBDTA-4</b>        |     |   |                                  |         |
| -1                         | --- | 1 | C (12) H (16) Ho (1) N (2) O (8) | 481.198 |

|                                |     |   |                                   |         |
|--------------------------------|-----|---|-----------------------------------|---------|
| <b>Ho+3_MeEDTA-4</b>           |     |   |                                   |         |
| -1                             | --- | 1 | C (11) H (14) Ho (1) N (2) O (8)  | 467.171 |
| <b>Ho+3_MeMalon-2</b>          |     |   |                                   |         |
| 1                              | --- | 2 | C (4) H (4) Ho (1) O (4)          | 281.003 |
| <b>Ho+3_MeMalon-2 (2)</b>      |     |   |                                   |         |
| -1                             | --- | 1 | C (8) H (8) Ho (1) O (8)          | 397.077 |
| <b>Ho+3_MeoxAcet-1</b>         |     |   |                                   |         |
| 2                              | --- | 2 | C (3) H (5) Ho (1) O (3)          | 254.001 |
| <b>Ho+3_MeoxAcet-1 (2)</b>     |     |   |                                   |         |
| 1                              | --- | 1 | C (6) H (10) Ho (1) O (6)         | 343.072 |
| <b>Ho+3_MIDA-2</b>             |     |   |                                   |         |
| 1                              | --- | 1 | C (5) H (7) Ho (1) N (1) O (4)    | 310.045 |
| <b>Ho+3_MIDA-2 (2)</b>         |     |   |                                   |         |
| -1                             | --- | 1 | C (10) H (14) Ho (1) N (2) O (8)  | 455.160 |
| <b>Ho+3_MoO4-2</b>             |     |   |                                   |         |
| 1                              | --- | 1 | Ho (1) Mo (1) O (4)               | 324.890 |
| <b>Ho+3_N23DiOHPrIDA-2</b>     |     |   |                                   |         |
| 1                              | --- | 1 | C (7) H (11) Ho (1) N (1) O (6)   | 370.097 |
| <b>Ho+3_N23DiOHPrIDA-2 (2)</b> |     |   |                                   |         |
| -1                             | --- | 1 | C (14) H (22) Ho (1) N (2) O (12) | 575.265 |
| <b>Ho+3_N2MeoxEtIDA-2</b>      |     |   |                                   |         |
| 1                              | --- | 1 | C (7) H (11) Ho (1) N (1) O (5)   | 354.098 |
| <b>Ho+3_N2MeoxEtIDA-2 (2)</b>  |     |   |                                   |         |
| -1                             | --- | 1 | C (14) H (22) Ho (1) N (2) O (10) | 543.266 |

|                                  |     |   |                             |         |
|----------------------------------|-----|---|-----------------------------|---------|
| <b>Ho+3_N2MeSEtIDA-2</b>         |     |   |                             |         |
| 1                                | --- | 1 | C(7)H(11)Ho(1)N(1)O(4)S(1)  | 370.159 |
| <b>Ho+3_N2MeSEtIDA-2(2)</b>      |     |   |                             |         |
| -1                               | --- | 1 | C(14)H(22)Ho(1)N(2)O(8)S(2) | 575.387 |
| <b>Ho+3_N2OHBenzIDA-3</b>        |     |   |                             |         |
| 0                                | --- | 1 | C(11)H(10)Ho(1)N(1)O(5)     | 401.134 |
| <b>Ho+3_N2OHBenzIDA-3(2)</b>     |     |   |                             |         |
| -3                               | --- | 1 | C(22)H(20)Ho(1)N(2)O(10)    | 637.338 |
| <b>Ho+3_N2PyridMeAsp-2</b>       |     |   |                             |         |
| 1                                | --- | 2 | C(10)H(10)Ho(1)N(2)O(4)     | 387.130 |
| <b>Ho+3_N2PyridMeAsp-2(2)</b>    |     |   |                             |         |
| -1                               | --- | 1 | C(20)H(20)Ho(1)N(4)O(8)     | 609.331 |
| <b>Ho+3_N6Me2PyridMeAsp-2</b>    |     |   |                             |         |
| 1                                | --- | 2 | C(11)H(12)Ho(1)N(2)O(4)     | 401.157 |
| <b>Ho+3_N6Me2PyridMeAsp-2(2)</b> |     |   |                             |         |
| -1                               | --- | 1 | C(22)H(24)Ho(1)N(4)O(8)     | 637.385 |
| <b>Ho+3_NAcAconylidOrth-2</b>    |     |   |                             |         |
| 1                                | --- | 1 | C(11)H(11)Ho(1)N(1)O(4)S(1) | 418.203 |
| <b>Ho+3_NBenzIDA-2</b>           |     |   |                             |         |
| 1                                | --- | 1 | C(11)H(11)Ho(1)N(1)O(4)     | 386.143 |
| <b>Ho+3_NBenzIDA-2(2)</b>        |     |   |                             |         |
| -1                               | --- | 1 | C(22)H(22)Ho(1)N(2)O(8)     | 607.355 |
| <b>Ho+3_Nicotinic-1</b>          |     |   |                             |         |
| 2                                | --- | 1 | C(6)H(4)Ho(1)N(1)O(2)       | 287.033 |



|                                 |     |   |                                         |         |
|---------------------------------|-----|---|-----------------------------------------|---------|
| <b>Ho+3_NMeEDTA-3</b>           |     |   |                                         |         |
| 0                               | --- | 1 | C (9) H (13) Ho (1) N (2) O (6)         | 410.142 |
| <b>Ho+3_NNBis2OHEtAla-1</b>     |     |   |                                         |         |
| 2                               | --- | 1 | C (7) H (14) Ho (1) N (1) O (4)         | 341.123 |
| <b>Ho+3_NNBis2OHEtAla-1 (2)</b> |     |   |                                         |         |
| 1                               | --- | 1 | C (14) H (28) Ho (1) N (2) O (8)        | 517.315 |
| <b>Ho+3_NO3-1</b>               |     |   |                                         |         |
| 2                               | --- | 1 | Ho (1) N (1) O (3)                      | 226.935 |
| <b>Ho+3_NONWAcid-2</b>          |     |   |                                         |         |
| 1                               | --- | 1 | C (10) H (5) Ho (1) N (1) O (5) S (1)   | 416.143 |
| <b>Ho+3_NTA-3</b>               |     |   |                                         |         |
| 0                               | --- | 5 | C (6) H (6) Ho (1) N (1) O (6)          | 353.047 |
| <b>Ho+3_NTA-3_EDTA-4</b>        |     |   |                                         |         |
| -4                              | --- | 2 | C (16) H (18) Ho (1) N (3) O (14)       | 641.261 |
| <b>Ho+3_NTA-3 (2)</b>           |     |   |                                         |         |
| -3                              | --- | 2 | C (12) H (12) Ho (1) N (2) O (12)       | 541.164 |
| <b>Ho+3_oAnisic-1</b>           |     |   |                                         |         |
| 2                               | --- | 1 | C (8) H (7) Ho (1) O (3)                | 316.072 |
| <b>Ho+3_oAnisic-1 (2)</b>       |     |   |                                         |         |
| 1                               | --- | 1 | C (16) H (14) Ho (1) O (6)              | 467.214 |
| <b>Ho+3_OH-1</b>                |     |   |                                         |         |
| 2                               | --- | 3 | H (1) Ho (1) O (1)                      | 181.937 |
| <b>Ho+3_OH-1_5ATP-4</b>         |     |   |                                         |         |
| -2                              | --- | 1 | C (10) H (13) Ho (1) N (5) O (14) P (3) | 685.090 |

|                                                                                                |     |   |                                         |         |
|------------------------------------------------------------------------------------------------|-----|---|-----------------------------------------|---------|
| <b>Ho+3_OH-1_HIMDA-2 (2)</b>                                                                   |     |   |                                         |         |
| -2                                                                                             | --- | 1 | C (12) H (19) Ho (1) N (2) O (11)       | 532.220 |
| <b>Ho+3_OH-1_HOEDTA-3</b>                                                                      |     |   |                                         |         |
| -1                                                                                             | --- | 1 | C (10) H (16) Ho (1) N (2) O (8)        | 457.176 |
| <b>Ho+3_OH-1_MeThymolBlue-6</b>                                                                |     |   |                                         |         |
| -4                                                                                             | --- | 1 | C (37) H (39) Ho (1) N (2) O (14) S (1) | 932.712 |
| <b>Ho+3_OH-1_NTA-3</b>                                                                         |     |   |                                         |         |
| -1                                                                                             | --- | 2 | C (6) H (7) Ho (1) N (1) O (7)          | 370.054 |
| <b>Ho+3_OH-1 (2)</b>                                                                           |     |   |                                         |         |
| 1                                                                                              | --- | 2 | H (2) Ho (1) O (2)                      | 198.945 |
| <b>Ho+3_OH-1 (3)</b>                                                                           |     |   |                                         |         |
| 0                                                                                              | --- | 2 | H (3) Ho (1) O (3)                      | 215.952 |
| <b>Ho+3_OH-1 (3) (am., s)</b><br>Holmium hydroxide, amorphous; Holmium trihydroxide, amorphous |     |   |                                         |         |
| 0                                                                                              | --- | 3 | H (3) Ho (1) O (3)                      | 215.952 |
| <b>Ho+3_OH-1 (3) (s)</b><br>Holmium hydroxide; Holmium trihydroxide                            |     |   |                                         |         |
| 0                                                                                              | --- | 3 | H (3) Ho (1) O (3)                      | 215.952 |
| <b>Ho+3_OH-1 (4)</b>                                                                           |     |   |                                         |         |
| -1                                                                                             | --- | 1 | H (4) Ho (1) O (4)                      | 232.959 |
| <b>Ho+3_Oxalic-2</b>                                                                           |     |   |                                         |         |
| 1                                                                                              | --- | 1 | C (2) Ho (1) O (4)                      | 252.950 |
| <b>Ho+3_Oxalic-2 (2)</b>                                                                       |     |   |                                         |         |
| -1                                                                                             | --- | 1 | C (4) Ho (1) O (8)                      | 340.969 |
| <b>Ho+3_Oxine-1</b>                                                                            |     |   |                                         |         |
| 2                                                                                              | --- | 2 | C (9) H (6) Ho (1) N (1) O (1)          | 309.083 |

|                            |     |   |                                   |         |
|----------------------------|-----|---|-----------------------------------|---------|
| <b>Ho+3_Oxine-1 (2)</b>    |     |   |                                   |         |
| 1                          | --- | 1 | C (18) H (12) Ho (1) N (2) O (2)  | 453.236 |
| <b>Ho+3_P3O10-5 (2)</b>    |     |   |                                   |         |
| -7                         | --- | 1 | Ho (1) O (20) P (6)               | 670.762 |
| <b>Ho+3_PAN-1</b>          |     |   |                                   |         |
| 2                          | --- | 2 | C (15) H (10) Ho (1) N (3) O (1)  | 413.194 |
| <b>Ho+3_PAN-1 (2)</b>      |     |   |                                   |         |
| 1                          | --- | 1 | C (30) H (20) Ho (1) N (6) O (2)  | 661.458 |
| <b>Ho+3_PAN-1 (3)</b>      |     |   |                                   |         |
| 0                          | --- | 1 | C (45) H (30) Ho (1) N (9) O (3)  | 909.722 |
| <b>Ho+3_PAN-1 (4)</b>      |     |   |                                   |         |
| -1                         | --- | 1 | C (60) H (40) Ho (1) N (12) O (4) | 1157.99 |
| <b>Ho+3_pAnisic-1</b>      |     |   |                                   |         |
| 2                          | --- | 1 | C (8) H (7) Ho (1) O (3)          | 316.072 |
| <b>Ho+3_PAR-2</b>          |     |   |                                   |         |
| 1                          | --- | 1 | C (11) H (7) Ho (1) N (3) O (2)   | 378.126 |
| <b>Ho+3_PhenAcet-1</b>     |     |   |                                   |         |
| 2                          | --- | 1 | C (8) H (7) Ho (1) O (2)          | 300.072 |
| <b>Ho+3_PhenEDTA-4</b>     |     |   |                                   |         |
| -1                         | --- | 1 | C (16) H (16) Ho (1) N (2) O (8)  | 529.242 |
| <b>Ho+3_Phthalic-2</b>     |     |   |                                   |         |
| 1                          | --- | 2 | C (8) H (4) Ho (1) O (4)          | 329.047 |
| <b>Ho+3_Phthalic-2 (2)</b> |     |   |                                   |         |
| -1                         | --- | 1 | C (16) H (8) Ho (1) O (8)         | 493.165 |

|                            |     |   |                         |         |
|----------------------------|-----|---|-------------------------|---------|
| <b>Ho+3_Picolinic-1</b>    |     |   |                         |         |
| 2                          | --- | 2 | C(6)H(4)Ho(1)N(1)O(2)   | 287.033 |
| <b>Ho+3_Picolinic-1(2)</b> |     |   |                         |         |
| 1                          | --- | 3 | C(12)H(8)Ho(1)N(2)O(4)  | 409.137 |
| <b>Ho+3_Picolinic-1(3)</b> |     |   |                         |         |
| 0                          | --- | 3 | C(18)H(12)Ho(1)N(3)O(6) | 531.240 |
| <b>Ho+3_Picolinic-1(4)</b> |     |   |                         |         |
| -1                         | --- | 1 | C(24)H(16)Ho(1)N(4)O(8) | 653.343 |
| <b>Ho+3_PO4-3</b>          |     |   |                         |         |
| 0                          | --- | 2 | Ho(1)O(4)P(1)           | 259.902 |
| <b>Ho+3_PO4-3_(s)</b>      |     |   |                         |         |
| 0                          | --- | 2 | Ho(1)O(4)P(1)           | 259.902 |
| <b>Ho+3_PO4-3_H2O_(s)</b>  |     |   |                         |         |
| 0                          | --- | 1 | H(2)Ho(1)O(5)P(1)       | 277.917 |
| <b>Ho+3_PO4-3(2)</b>       |     |   |                         |         |
| -3                         | --- | 2 | Ho(1)O(8)P(2)           | 354.873 |
| <b>Ho+3_Pro-1</b>          |     |   |                         |         |
| 2                          | --- | 1 | C(5)H(8)Ho(1)N(1)O(2)   | 279.054 |
| <b>Ho+3_ProGly-1</b>       |     |   |                         |         |
| 2                          | --- | 1 | C(7)H(11)Ho(1)N(2)O(3)  | 336.106 |
| <b>Ho+3_Propanoic-1</b>    |     |   |                         |         |
| 2                          | --- | 2 | C(3)H(5)Ho(1)O(2)       | 238.002 |
| <b>Ho+3_Propanoic-1(2)</b> |     |   |                         |         |
| 1                          | --- | 2 | C(6)H(10)Ho(1)O(4)      | 311.073 |

|                            |     |   |                      |         |
|----------------------------|-----|---|----------------------|---------|
| <b>Ho+3_Propanoic-1(3)</b> |     |   |                      |         |
| 0                          | --- | 1 | C(9)H(15)Ho(1)O(6)   | 384.145 |
| <b>Ho+3_Propanoic-1(4)</b> |     |   |                      |         |
| -1                         | --- | 1 | C(12)H(20)Ho(1)O(8)  | 457.216 |
| <b>Ho+3_Pyruvic-1</b>      |     |   |                      |         |
| 2                          | --- | 1 | C(3)H(3)Ho(1)O(3)    | 251.985 |
| <b>Ho+3_Quinic-1</b>       |     |   |                      |         |
| 2                          | --- | 2 | C(7)H(11)Ho(1)O(6)   | 356.091 |
| <b>Ho+3_Quinic-1(2)</b>    |     |   |                      |         |
| 1                          | --- | 2 | C(14)H(22)Ho(1)O(12) | 547.252 |
| <b>Ho+3_Quinic-1(3)</b>    |     |   |                      |         |
| 0                          | --- | 2 | C(21)H(33)Ho(1)O(18) | 738.412 |
| <b>Ho+3_Quinic-1(4)</b>    |     |   |                      |         |
| -1                         | --- | 1 | C(28)H(44)Ho(1)O(24) | 929.573 |
| <b>Ho+3_Salicylic-2</b>    |     |   |                      |         |
| 1                          | --- | 2 | C(7)H(4)Ho(1)O(3)    | 301.037 |
| <b>Ho+3_Salicylic-2(2)</b> |     |   |                      |         |
| -1                         | --- | 3 | C(14)H(8)Ho(1)O(6)   | 437.144 |
| <b>Ho+3_Salicylic-2(3)</b> |     |   |                      |         |
| -3                         | --- | 1 | C(21)H(12)Ho(1)O(9)  | 573.251 |
| <b>Ho+3_SCN-1</b>          |     |   |                      |         |
| 2                          | --- | 2 | C(1)Ho(1)N(1)S(1)    | 223.008 |
| <b>Ho+3_SCN-1(2)</b>       |     |   |                      |         |
| 1                          | --- | 2 | C(2)Ho(1)N(2)S(2)    | 281.085 |

|                             |     |   |                                         |         |
|-----------------------------|-----|---|-----------------------------------------|---------|
| <b>Ho+3_SCN-1 (3)</b>       |     |   |                                         |         |
| 0                           | --- | 1 | C (3) Ho (1) N (3) S (3)                | 339.163 |
| <b>Ho+3_Ser-1</b>           |     |   |                                         |         |
| 2                           | --- | 1 | C (3) H (6) Ho (1) N (1) O (3)          | 269.016 |
| <b>Ho+3_SO4-2</b>           |     |   |                                         |         |
| 1                           | --- | 3 | Ho (1) O (4) S (1)                      | 260.988 |
| <b>Ho+3_SO4-2 (2)</b>       |     |   |                                         |         |
| -1                          | --- | 2 | Ho (1) O (8) S (2)                      | 357.045 |
| <b>Ho+3_Sulfox-2_EDTA-4</b> |     |   |                                         |         |
| -3                          | --- | 2 | C (19) H (17) Ho (1) N (3) O (12) S (1) | 676.347 |
| <b>Ho+3_SulfSal-3</b>       |     |   |                                         |         |
| 0                           | --- | 2 | C (7) H (3) Ho (1) O (6) S (1)          | 380.087 |
| <b>Ho+3_SulfSal-3 (2)</b>   |     |   |                                         |         |
| -3                          | --- | 1 | C (14) H (6) Ho (1) O (12) S (2)        | 595.244 |
| <b>Ho+3_TAR-2</b>           |     |   |                                         |         |
| 1                           | --- | 1 | C (9) H (5) Ho (1) N (3) O (2) S (1)    | 384.148 |
| <b>Ho+3_Tartaric-2</b>      |     |   |                                         |         |
| 1                           | --- | 1 | C (4) H (4) Ho (1) O (6)                | 313.002 |
| <b>Ho+3_TEDTA-4</b>         |     |   |                                         |         |
| -1                          | --- | 1 | C (12) H (16) Ho (1) N (2) O (8) S (1)  | 513.258 |
| <b>Ho+3_TetMDTA-4</b>       |     |   |                                         |         |
| -1                          | --- | 2 | C (12) H (16) Ho (1) N (2) O (8)        | 481.198 |
| <b>Ho+3_Tetraglyme</b>      |     |   |                                         |         |
| 3                           | --- | 1 | C (10) H (22) Ho (1) O (5)              | 387.212 |

|                             |     |   |                                   |         |
|-----------------------------|-----|---|-----------------------------------|---------|
| <b>Ho+3_Tiron-4</b>         |     |   |                                   |         |
| -1                          | --- | 1 | C (6) H (2) Ho (1) O (8) S (2)    | 431.127 |
| <b>Ho+3_Tiron-4 (2)</b>     |     |   |                                   |         |
| -5                          | --- | 1 | C (12) H (4) Ho (1) O (16) S (4)  | 697.324 |
| <b>Ho+3_Trien</b>           |     |   |                                   |         |
| 3                           | --- | 1 | C (6) H (18) Ho (1) N (4)         | 311.166 |
| <b>Ho+3_Trien (2)</b>       |     |   |                                   |         |
| 3                           | --- | 1 | C (12) H (36) Ho (1) N (8)        | 457.401 |
| <b>Ho+3_TriMDTA-4</b>       |     |   |                                   |         |
| -1                          | --- | 1 | C (11) H (14) Ho (1) N (2) O (8)  | 467.171 |
| <b>Ho+3_Tropic-1</b>        |     |   |                                   |         |
| 2                           | --- | 1 | C (9) H (9) Ho (1) O (3)          | 330.099 |
| <b>Ho+3_Tropic-1 (2)</b>    |     |   |                                   |         |
| 1                           | --- | 1 | C (18) H (18) Ho (1) O (6)        | 495.267 |
| <b>Ho+3_Tropolone-1</b>     |     |   |                                   |         |
| 2                           | --- | 2 | C (7) H (5) Ho (1) O (2)          | 286.046 |
| <b>Ho+3_Tropolone-1 (2)</b> |     |   |                                   |         |
| 1                           | --- | 2 | C (14) H (10) Ho (1) O (4)        | 407.161 |
| <b>Ho+3_Tropolone-1 (3)</b> |     |   |                                   |         |
| 0                           | --- | 2 | C (21) H (15) Ho (1) O (6)        | 528.277 |
| <b>Ho+3_Tropolone-1 (4)</b> |     |   |                                   |         |
| -1                          | --- | 1 | C (28) H (20) Ho (1) O (8)        | 649.392 |
| <b>Ho+3_TTHA-6</b>          |     |   |                                   |         |
| -3                          | --- | 2 | C (18) H (24) Ho (1) N (4) O (12) | 653.338 |

|                                                                                                                                                 |            |     |                                   |         |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|-----------------------------------|---------|
| <b>Ho+3_Val-1</b>                                                                                                                               |            |     |                                   |         |
| 2                                                                                                                                               | ---        | 1   | C (5) H (10) Ho (1) N (1) O (2)   | 281.070 |
| <b>Ho+3 (2)_CO3-2 (3)_ (s)</b>                                                                                                                  |            |     |                                   |         |
| 0                                                                                                                                               | ---        | 1   | C (3) Ho (2) O (9)                | 509.888 |
| <b>Ho+3 (2)_EDTA-4 (3)</b>                                                                                                                      |            |     |                                   |         |
| -6                                                                                                                                              | ---        | 2   | C (30) H (36) Ho (2) N (6) O (24) | 1194.50 |
| <b>Ho+3 (2)_Na+1 (2)_SO4-2 (4)_H2O (2)_ (s)</b>                                                                                                 |            |     |                                   |         |
| 0                                                                                                                                               | ---        | 1   | H (4) Ho (2) Na (2) O (18) S (4)  | 796.101 |
| <b>Ho+3 (2)_P2O7-4</b>                                                                                                                          |            |     |                                   |         |
| 2                                                                                                                                               | ---        | 1   | Ho (2) O (7) P (2)                | 503.804 |
| <b>Ho+3 (2)_SO4-2 (3)_H2O (8)_ (s)</b><br>Holmium(III) sulfate octahydrate                                                                      |            |     |                                   |         |
| 0                                                                                                                                               | ---        | 1   | H (16) Ho (2) O (20) S (3)        | 762.155 |
| <b>Ho+3 (2)_TTHA-6</b>                                                                                                                          |            |     |                                   |         |
| 0                                                                                                                                               | ---        | 1   | C (18) H (24) Ho (2) N (4) O (12) | 818.268 |
| <b>Ho (s)</b><br>Holmium                                                                                                                        |            |     |                                   |         |
| 0                                                                                                                                               | ---        | 13  | Ho (1)                            | 164.930 |
| <b>Ho2O3 (s)</b><br>Holmium oxide                                                                                                               |            |     |                                   |         |
| 0                                                                                                                                               | ---        | 2   | Ho (2) O (3)                      | 377.858 |
| <b>HOEDTA-3</b><br>N-[2-[bis(carboxymethyl)amino]ethyl]-N-(2-hydroxyethyl)glycinate ion; N-(2-Hydroxyethyl)ethylenedinitrilotriacetate; HEDTA-3 |            |     |                                   |         |
| -3                                                                                                                                              | 14047-40-6 | 321 | C (10) H (15) N (2) O (7)         | 275.238 |
| <b>HydrazineNNDiAcet-2</b><br>Hydrazine-N,N'-diacetate anion                                                                                    |            |     |                                   |         |
| -2                                                                                                                                              | ---        | 80  | C (4) H (6) N (2) O (4)           | 146.103 |



|                                                                                                                                                   |            |     |                    |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|--------------------|---------|
| <b>Hyp-1</b><br>Hydroxyprolinate ion; L-Hydroxyprolinate ion; 4-hydroxy-2-pyrrolidinecarboxylate ion;<br>L-4-hydroxypyrrolidine-2-carboxylate ion |            |     |                    |         |
| -1                                                                                                                                                | ---        | 402 | C(5)H(8)N(1)O(3)   | 130.123 |
| <b>IButanoic-1</b><br>Isobutanoate ion; Isobutyrate ion                                                                                           |            |     |                    |         |
| -1                                                                                                                                                | ---        | 75  | C(4)H(7)O(2)       | 87.0984 |
| <b>IDA-2</b><br>Iminodiacetate ion                                                                                                                |            |     |                    |         |
| -2                                                                                                                                                | 28528-43-0 | 428 | C(4)H(5)N(1)O(4)   | 131.088 |
| <b>IDS-4</b><br>Iminodisuccinate anion                                                                                                            |            |     |                    |         |
| -4                                                                                                                                                | ---        | 51  | C(8)H(7)N(1)O(8)   | 245.146 |
| <b>IO3-1</b><br>Iodate ion                                                                                                                        |            |     |                    |         |
| -1                                                                                                                                                | 15454-31-6 | 291 | I(1)O(3)           | 174.898 |
| <b>Itaconic-2</b><br>Itaconate ion; Methylenesuccinate ion                                                                                        |            |     |                    |         |
| -2                                                                                                                                                | ---        | 61  | C(5)H(4)O(4)       | 128.084 |
| <b>Lactic-1</b><br>Lactate ion                                                                                                                    |            |     |                    |         |
| -1                                                                                                                                                | ---        | 643 | C(3)H(5)O(3)       | 89.0709 |
| <b>LeuGly-1</b>                                                                                                                                   |            |     |                    |         |
| -1                                                                                                                                                | ---        | 49  | C(8)H(15)N(2)O(3)  | 187.219 |
| <b>LeuGlyGly-1</b>                                                                                                                                |            |     |                    |         |
| -1                                                                                                                                                | ---        | 23  | C(10)H(18)N(3)O(4) | 244.271 |
| <b>Maleic-2</b><br>cis-Butenedioate ion; Maleate ion                                                                                              |            |     |                    |         |
| -2                                                                                                                                                | 142-44-9   | 195 | C(4)H(2)O(4)       | 114.058 |

|                                                                                                                                                                                       |            |     |                    |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|--------------------|---------|
| <b>Malic-2</b><br>Malate ion                                                                                                                                                          |            |     |                    |         |
| -2                                                                                                                                                                                    | 149-61-1   | 682 | C(4)H(4)O(5)       | 132.073 |
| <b>Malonic-2</b><br>Malonate ion                                                                                                                                                      |            |     |                    |         |
| -2                                                                                                                                                                                    | 156-80-9   | 417 | C(3)H(2)O(4)       | 102.047 |
| <b>Mandelic-1</b><br>alpha-Hydroxybenzeneacetate ion; Mandelate ion; L-Phenylhydroxyacetate ion;<br>Hydroxybenzene acetate ion; Phenylglycolate ion; Uromalinate ion; Amaygdalate ion |            |     |                    |         |
| -1                                                                                                                                                                                    | 769-61-9   | 132 | C(8)H(7)O(3)       | 151.142 |
| <b>mAnisic-1</b>                                                                                                                                                                      |            |     |                    |         |
| -1                                                                                                                                                                                    | ---        | 15  | C(8)H(7)O(3)       | 151.142 |
| <b>mBDTA-4</b><br>meso-2,3-Butylenedinitrilotetraacetate ion; meso-1,2-Dimethylethylenedinitrilotetraacetate ion; mBDTA anion                                                         |            |     |                    |         |
| -4                                                                                                                                                                                    | ---        | 58  | C(12)H(16)N(2)O(8) | 316.268 |
| <b>MeEDTA-4</b>                                                                                                                                                                       |            |     |                    |         |
| -4                                                                                                                                                                                    | ---        | 57  | C(11)H(14)N(2)O(8) | 302.241 |
| <b>MeMalon-2</b>                                                                                                                                                                      |            |     |                    |         |
| -2                                                                                                                                                                                    | ---        | 55  | C(4)H(4)O(4)       | 116.073 |
| <b>MeoxAcet-1</b>                                                                                                                                                                     |            |     |                    |         |
| -1                                                                                                                                                                                    | ---        | 37  | C(3)H(5)O(3)       | 89.0709 |
| <b>MIDA-2</b>                                                                                                                                                                         |            |     |                    |         |
| -2                                                                                                                                                                                    | ---        | 147 | C(5)H(7)N(1)O(4)   | 145.115 |
| <b>MoO4-2</b><br>Molybdate(VI) ion                                                                                                                                                    |            |     |                    |         |
| -2                                                                                                                                                                                    | 14259-85-9 | 281 | Mo(1)O(4)          | 159.960 |
| <b>N23DiOHPrIDA-2</b>                                                                                                                                                                 |            |     |                    |         |
| -2                                                                                                                                                                                    | ---        | 44  | C(7)H(11)N(1)O(6)  | 205.167 |

|                                                                                                                                                                                                                  |           |     |                        |         |
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| <b>N2MeoxEtIDA-2</b><br>N-(2-Methoxyethyl)iminodiacetic acid                                                                                                                                                     |           |     |                        |         |
| -2                                                                                                                                                                                                               | ---       | 65  | C(7)H(11)N(1)O(5)      | 189.168 |
| <b>N2MeSEtIDA-2</b>                                                                                                                                                                                              |           |     |                        |         |
| -2                                                                                                                                                                                                               | ---       | 59  | C(7)H(11)N(1)O(4)S(1)  | 205.229 |
| <b>N2OHBenzIDA-3</b><br>N-(2-Hydroxybenzyl)iminodiacetate anion; N-(2-Hydroxybenzyl)-N-(carboxymethyl)glycinate anion; N-(Carboxymethyl)-N-(2-hydroxybenzyl)glycinate anion; N-Hydroxybenzyliminodiacetate anion |           |     |                        |         |
| -3                                                                                                                                                                                                               | ---       | 88  | C(11)H(10)N(1)O(5)     | 236.204 |
| <b>N2PyridMeAsp-2</b><br>N-(2-Pyridylmethyl)-L-aspartate ion; N-(2-Pyridylmethyl)iminodiacetate ion                                                                                                              |           |     |                        |         |
| -2                                                                                                                                                                                                               | ---       | 88  | C(10)H(10)N(2)O(4)     | 222.200 |
| <b>N6Me2PyridMeAsp-2</b><br>N-(6-Methyl-2-pyridylmethyl)iminodiacetate ion; N-(6-Methyl-2-pyridylmethyl)-L-aspartate ion                                                                                         |           |     |                        |         |
| -2                                                                                                                                                                                                               | ---       | 67  | C(11)H(12)N(2)O(4)     | 236.227 |
| <b>Na(s)</b><br>Sodium; Sodium, cubic                                                                                                                                                                            |           |     |                        |         |
| 0                                                                                                                                                                                                                | 7440-23-5 | 209 | Na(1)                  | 22.9900 |
| <b>NAcAconylidOrth-2</b>                                                                                                                                                                                         |           |     |                        |         |
| -2                                                                                                                                                                                                               | ---       | 11  | C(11)H(11)N(1)O(4)S(1) | 253.273 |
| <b>NBenzIDA-2</b>                                                                                                                                                                                                |           |     |                        |         |
| -2                                                                                                                                                                                                               | ---       | 53  | C(11)H(11)N(1)O(4)     | 221.213 |
| <b>Nicotinic-1</b><br>Nicotinate ion                                                                                                                                                                             |           |     |                        |         |
| -1                                                                                                                                                                                                               | ---       | 39  | C(6)H(4)N(1)O(2)       | 122.103 |
| <b>NMeEDTA-3</b><br>Methyl-N-ethylenediaminetriacetate ion                                                                                                                                                       |           |     |                        |         |
| -3                                                                                                                                                                                                               | ---       | 20  | C(9)H(13)N(2)O(6)      | 245.212 |

|                                                                                                                            |            |      |                       |         |
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| <b>NNBis2OHEtAla-1</b>                                                                                                     |            |      |                       |         |
| -1                                                                                                                         | ---        | 40   | C(7)H(14)N(1)O(4)     | 176.193 |
| <b>NO3-1</b><br>Nitrate ion                                                                                                |            |      |                       |         |
| -1                                                                                                                         | 14797-55-8 | 573  | N(1)O(3)              | 62.0049 |
| <b>NONWAcid-2</b>                                                                                                          |            |      |                       |         |
| -2                                                                                                                         | ---        | 26   | C(10)H(5)N(1)O(5)S(1) | 251.213 |
| <b>NTA-3</b><br>Nitrilotriacetate ion; N,N-bis(carboxymethyl)glycinate ion; Triglycollamate ion; Triglycinate ion; NTA ion |            |      |                       |         |
| -3                                                                                                                         | 28528-44-1 | 751  | C(6)H(6)N(1)O(6)      | 188.117 |
| <b>O2</b><br>Oxygen (aquated)                                                                                              |            |      |                       |         |
| 0                                                                                                                          | ---        | 67   | O(2)                  | 31.9988 |
| <b>O2 (g)</b><br>Oxygen gas; Oxygen                                                                                        |            |      |                       |         |
| 0                                                                                                                          | 7782-44-7  | 2559 | O(2)                  | 31.9988 |
| <b>oAnisic-1</b>                                                                                                           |            |      |                       |         |
| -1                                                                                                                         | ---        | 35   | C(8)H(7)O(3)          | 151.142 |
| <b>OH-1</b><br>Hydroxide ion                                                                                               |            |      |                       |         |
| -1                                                                                                                         | 14280-30-9 | 6558 | H(1)O(1)              | 17.0073 |
| <b>Oxalic-2</b><br>Oxalate ion                                                                                             |            |      |                       |         |
| -2                                                                                                                         | 338-70-5   | 1241 | C(2)O(4)              | 88.0196 |
| <b>Oxine-1</b><br>8-Hydroxyquinolate ion; Oxine ion                                                                        |            |      |                       |         |
| -1                                                                                                                         | 16582-16-4 | 162  | C(9)H(6)N(1)O(1)      | 144.153 |
| <b>P2O7-4</b><br>Pyrophosphate ion; Pyrophosphate anion; Diphosphate ion                                                   |            |      |                       |         |

|                                                                                                               |            |      |                    |         |
|---------------------------------------------------------------------------------------------------------------|------------|------|--------------------|---------|
| -4                                                                                                            | 14000-31-8 | 285  | O(7)P(2)           | 173.944 |
| <b>P3O10-5</b><br>Triphosphate ion; Tripolyphosphate ion                                                      |            |      |                    |         |
| -5                                                                                                            | 14127-68-5 | 283  | O(10)P(3)          | 252.916 |
| <b>PAN-1</b>                                                                                                  |            |      |                    |         |
| -1                                                                                                            | ---        | 38   | C(15)H(10)N(3)O(1) | 248.264 |
| <b>pAnisic-1</b>                                                                                              |            |      |                    |         |
| -1                                                                                                            | ---        | 15   | C(8)H(7)O(3)       | 151.142 |
| <b>PAR-2</b><br>Pyridylazoresorcinolate ion                                                                   |            |      |                    |         |
| -2                                                                                                            | ---        | 110  | C(11)H(7)N(3)O(2)  | 213.196 |
| <b>PhenAcet-1</b><br>Phenylacetate ion; 2-Phenylacetate                                                       |            |      |                    |         |
| -1                                                                                                            | ---        | 36   | C(8)H(7)O(2)       | 135.142 |
| <b>PhenEDTA-4</b><br>Phenylethylenediaminetetraacetate ion                                                    |            |      |                    |         |
| -4                                                                                                            | ---        | 30   | C(16)H(16)N(2)O(8) | 364.312 |
| <b>Phthalic-2</b><br>Phthalate ion                                                                            |            |      |                    |         |
| -2                                                                                                            | 3198-29-6  | 134  | C(8)H(4)O(4)       | 164.117 |
| <b>Picolinic-1</b><br>Picolinate ion                                                                          |            |      |                    |         |
| -1                                                                                                            | 3198-27-4  | 208  | C(6)H(4)N(1)O(2)   | 122.103 |
| <b>PO4-3</b><br>Phosphate ion; Phosphate ion                                                                  |            |      |                    |         |
| -3                                                                                                            | 14265-44-2 | 1594 | O(4)P(1)           | 94.9716 |
| <b>Pro-1</b><br>Prolinate ion; L-Prolinate ion; 2-pyrrolidinecarboxylate ion; L-pyrrolidine-2-carboxylate ion |            |      |                    |         |
| -1                                                                                                            | 17781-82-7 | 494  | C(5)H(8)N(1)O(2)   | 114.124 |

|                                                                                                                                                                                                                    |            |      |                   |         |
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| <b>ProGly-1</b>                                                                                                                                                                                                    |            |      |                   |         |
| -1                                                                                                                                                                                                                 | ---        | 40   | C(7)H(11)N(2)O(3) | 171.176 |
| <b>Propanoic-1</b><br>Propanoate ion; Propionate ion                                                                                                                                                               |            |      |                   |         |
| -1                                                                                                                                                                                                                 | 72-03-7    | 176  | C(3)H(5)O(2)      | 73.0715 |
| <b>Pyruvic-1</b><br>Pyruvate ion                                                                                                                                                                                   |            |      |                   |         |
| -1                                                                                                                                                                                                                 | 57-60-3    | 296  | C(3)H(3)O(3)      | 87.0550 |
| <b>Quinic-1</b><br>Quinate ion                                                                                                                                                                                     |            |      |                   |         |
| -1                                                                                                                                                                                                                 | 7729-33-1  | 44   | C(7)H(11)O(6)     | 191.161 |
| <b>S(s)</b><br>Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb                                                                                            |            |      |                   |         |
| 0                                                                                                                                                                                                                  | 7704-34-9  | 619  | S(1)              | 32.0600 |
| <b>Salicylic-2</b><br>Salicylate ion                                                                                                                                                                               |            |      |                   |         |
| -2                                                                                                                                                                                                                 | 16887-56-2 | 558  | C(7)H(4)O(3)      | 136.107 |
| <b>SCN-1</b><br>Thiocyanate ion; NCS-1 equivalent                                                                                                                                                                  |            |      |                   |         |
| -1                                                                                                                                                                                                                 | 302-04-5   | 785  | C(1)N(1)S(1)      | 58.0777 |
| <b>Ser-1</b><br>Serinate ion; L-Serinate ion; 2-amino-3-hydroxypropionate ion; beta-hydroxyalanate ion; 2-amino-3-hydroxypropanoate ion; alpha-amino-beta-hydroxypropionate ion; L-2-amino-3-hydroxypropanoate ion |            |      |                   |         |
| -1                                                                                                                                                                                                                 | 17807-54-4 | 708  | C(3)H(6)N(1)O(3)  | 104.086 |
| <b>SO4-2</b><br>Sulfate ion; Sulphate ion                                                                                                                                                                          |            |      |                   |         |
| -2                                                                                                                                                                                                                 | 14808-79-8 | 1270 | O(4)S(1)          | 96.0576 |
| <b>Sulfox-2</b><br>8-Hydroxy-5-quinoline sulphonate ion; 8-Hydroxy-5-quinoline sulfonate ion; 8-Hydroxyquinoline-5-sulfonate ion; 8-Hydroxyquinoline-5-sulphonate ion                                              |            |      |                   |         |

|                                                                                                                                                                                                                                                                                                                    |            |     |                        |         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|------------------------|---------|
| -2                                                                                                                                                                                                                                                                                                                 | 40747-73-7 | 190 | C(9)H(5)N(1)O(4)S(1)   | 223.203 |
| <b>SulfSal-3</b><br>5-Sulfosalicylate ion                                                                                                                                                                                                                                                                          |            |     |                        |         |
| -3                                                                                                                                                                                                                                                                                                                 | ---        | 238 | C(7)H(3)O(6)S(1)       | 215.157 |
| <b>TAR-2</b>                                                                                                                                                                                                                                                                                                       |            |     |                        |         |
| -2                                                                                                                                                                                                                                                                                                                 | ---        | 48  | C(9)H(5)N(3)O(2)S(1)   | 219.218 |
| <b>Tartaric-2</b><br>Tartrate ion                                                                                                                                                                                                                                                                                  |            |     |                        |         |
| -2                                                                                                                                                                                                                                                                                                                 | ---        | 393 | C(4)H(4)O(6)           | 148.072 |
| <b>TEDTA-4</b><br>Thiobis(ethylenenitrilo)tetraacetate ion; 2,2'-Diaminodiethylsulphide-NNN'N'-tetraacetic acid                                                                                                                                                                                                    |            |     |                        |         |
| -4                                                                                                                                                                                                                                                                                                                 | ---        | 63  | C(12)H(16)N(2)O(8)S(1) | 348.328 |
| <b>TetMDTA-4</b><br>Tetramethylenedinitrilotetraacetate                                                                                                                                                                                                                                                            |            |     |                        |         |
| -4                                                                                                                                                                                                                                                                                                                 | ---        | 103 | C(12)H(16)N(2)O(8)     | 316.268 |
| <b>Tetraglyme</b><br>Tetraglyme; 2,5,8,11,14-Pentaoxapentadecane; Tetraethylene glycol dimethyl ether; Dimethoxytetraethyleneglycol                                                                                                                                                                                |            |     |                        |         |
| 0                                                                                                                                                                                                                                                                                                                  | 143-24-8   | 22  | C(10)H(22)O(5)         | 222.282 |
| <b>Tiron-4</b><br>4,5-Dihydroxy-1,3-benzenedisulphonate ion; 4,5-Dihydroxy-1,3-benzenedisulfonate ion; 1,2-Dihydroxybenzene-3,5-disulphonate ion; 4,5-Dihydroxybenzene-1,3-disulphonate ion; Disodium 3,5-pyrocatecholdisulfonate ion; PDS                                                                         |            |     |                        |         |
| -4                                                                                                                                                                                                                                                                                                                 | 77310-82-8 | 263 | C(6)H(2)O(8)S(2)       | 266.197 |
| <b>Trien</b><br>Triethylenetetramine; 1,4,7,10-Tetraazadecane; N,N'-bis(2-aminoethyl)-1,2-ethanediamine; Trien; 2,2,2-tet; 3,6-Diazaoctane-1,8-diamine; Ethylenebis(2'-aminoethylamine); TETA (Triethylenetetramine); 3,6NH-od; 1,2-Diamino-N,N'-di(2-aminoethyl)ethane; N1,N2-Bis(2-aminoethyl)-1,2-ethanediamine |            |     |                        |         |
| 0                                                                                                                                                                                                                                                                                                                  | 112-24-3   | 155 | C(6)H(18)N(4)          | 146.236 |
| <b>TriMDTA-4</b>                                                                                                                                                                                                                                                                                                   |            |     |                        |         |
| -4                                                                                                                                                                                                                                                                                                                 | ---        | 107 | C(11)H(14)N(2)O(8)     | 302.241 |

|                                                                                    |            |     |                            |         |
|------------------------------------------------------------------------------------|------------|-----|----------------------------|---------|
| <b>Tropic-1</b>                                                                    |            |     |                            |         |
| -1                                                                                 | ---        | 27  | C (9) H (9) O (3)          | 165.169 |
| <b>Tropolone-1</b><br>Tropolonate ion; 1-Hydroxycyclohepta-3,5,7-trien-2-onate ion |            |     |                            |         |
| -1                                                                                 | 28318-47-0 | 110 | C (7) H (5) O (2)          | 121.116 |
| <b>TTHA-6</b><br>Triethylenetetranitrilohexa-acetate                               |            |     |                            |         |
| -6                                                                                 | ---        | 266 | C (18) H (24) N (4) O (12) | 488.408 |
| <b>Val-1</b><br>L-Valinate ion; Valinate ion; L-2-Amino-3-methylbutanoate ion      |            |     |                            |         |
| -1                                                                                 | ---        | 582 | C (5) H (10) N (1) O (2)   | 116.140 |